

TOP

100

HTML



CSS



BOOTSTRAP



INTERVIEW QUESTIONS

HAPPY RAWAT

PREFACE

ABOUT THE BOOK

This book contains 200 very important JavaScript interview questions.



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Top 100 HTML, CSS, Bootstrap Interview Q&A

HTML

1. [Basics & Document Structure](#)

2. HTML Elements & Tags – Part I

3. HTML Elements & Tags – Part II

4. HTML Attributes

5. HTML Links & Navigation

6. HTML Lists & Tables

7. HTML Images & Multimedia

8. HTML Forms

9. Best Practices & Deployment

10. Responsiveness & Others

CSS

1. Basics & Implementation

2. Selectors in CSS

3. Layout and Positioning

4. Responsive Design

Bootstrap

1. Bootstrap

1. HTML Basics & Document Structure

Q. What is **HTML**? Differences between **HTML** & **HTML5**? **Advantages** of HTML5?

Q. What is the difference between **HTML** and **XHTML**?

Q. What is the role of **DOCTYPE** in HTML?

Q. What if you remove `<!DOCTYPE html>` from your HTML?

Q. Difference between **Head** & **Body** in HTML? Where to place **JS link** reference?

Q. What is **Title Tag** in HTML? What are the **4 advantages** of Title tag?

Q. What are **Meta Tags**? What are the **5 types** of meta tags?

Q. What is **HTML**? Differences between **HTML** & **HTML5**? **Advantages** of HTML5? **V. IMP.**



- ❖ HTML(HyperText Markup Language) is the standard **markup language** used to create web pages.
- ❖ Markup language meaning a language which define the **structure of a document using elements** like headings, paragraphs, links, lists, and more.
- ❖ **HTML is not a programming language**, it's a markup language like XML.

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```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML</title>
  </head>
  <body>
    <a href="https://abc.com">Link</a>
    <h1>Heading Text</h1>
    <p>Para Text</p>
  </body>
</html>
```

Q. What is **HTML**? Differences between **HTML** & **HTML5**? **Advantages** of HTML5? **V. IMP.**



❖ HTML5 is the fifth and **latest version** of HTML.

Old HTML/ HTML4

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01//EN"
"http://www.w3.org/TR/html4/strict.dtd">
<html>
  <head>
    <title>Old HTML Example</title>
    <meta http-equiv="Content-Type"
    content="text/html; charset=UTF-8" />
  </head>
  <body>
    <h1>Tag</h1>
    <p>Paragraph</p>
    <a href="https://abc.com">Link</a>
  </body>
</html>
```

HTML/ HTML5

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>HTML5 Example</title>
  </head>
  <body>
    <header>
      <h1>Tag</h1>
    </header>
    <section>
      <p>Paragraph</p>
      <a href="https://abc.com">Link</a>
    </section>
  </body>
</html>
```

Q. What is **HTML**? Differences between **HTML** & **HTML5**? **Advantages** of HTML5? **V. IMP.**



5 Advantages of HTML5:

1. New Semantic Elements: `<header>`, `<nav>`, `<article>`, `<section>`, `<aside>`, `<footer>`

2. Form Input Types: `<input type="date">`, `<input type="email">`

3. Audio and Video Support: `<audio>`, `<video>`

4. Mobile Compatibility

5. Simpler Code

Q. What is the difference between HTML and XHTML?



- ❖ XHTML(eXtensible HyperText Markup Language) is a markup language that follows the **rules of XML** to define the structure of web pages.

XHTML

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE html
PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>XHTML Example</title>
  </head>
  <body>
    <h1>XHTML</h1>
    
  </body>
</html>
```

HTML/ HTML5

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML</title>
  </head>
  <body>
    <h1>HTML</h1>
    
  </body>
</html>
```

Q. What is the difference between **HTML** and **XHTML**?



HTML/ HTML5	XHTML
1. HTML has more lenient syntax rules. Closing tag is not mandatory for some elements. <code></code> .	XHTML follows stricter syntax rules. All tags must be properly nested and closed. <code></code>
2. HTML is not case-sensitive . <code><DIV></code> , <code><P></code> will work.	XHTML is case-sensitive. <code><div></code> , <code><p></code> will work.
3. HTML is widely supported by all browsers and web platforms.	XHTML has limited support by browsers.

Q. What is the difference between **HTML** and **XHTML**?



HTML/ HTML5	XHTML
1. HTML has more lenient syntax rules. Closing tag is not mandatory for some elements. <code></code> .	XHTML follows stricter syntax rules. All tags must be properly nested and closed. <code></code>
2. HTML is not case-sensitive . <code><DIV></code> , <code><P></code> will work.	XHTML is case-sensitive. <code><div></code> , <code><p></code> will work.
3. HTML is widely supported by all browsers and web platforms.	XHTML has limited support by browsers.

Q. What is the role of DOCTYPE in HTML? V. IMP.



- ❖ DOCTYPE(Document Type) declaration specifies the **version** of HTML.
- ❖ DOCTYPE tells the browser which version of HTML it is and how to interpret the code.

```
<!DOCTYPE html>
<html>
  <head>
    <title>DocType</title>
  </head>
  <body>
    <h1>DocType</h1>
  </body>
</html>
```

```
<!-- Different kind of Doctypes -->
<!-- HTML 4.01 Strict -->
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtdhttp://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd
```

Q. What if you **remove** `<!DOCTYPE html>` from your HTML?



- ❖ Then browsers can still render the page, but they will not be able to validate the version of HTML, therefore it may lead to some **compatibility issues** with SEO or debugging.

```
<!DOCTYPE html>
<html>
  <head>
    <title>DocType</title>
  </head>
  <body>
    <h1>DocType</h1>
  </body>
</html>
```

Q. Difference between **Head** & **Body** in HTML? Where to place **JS link** reference? **V. IMP.**



❖ The <head> element is where you place **meta-information** (information about the document). For example, <title>, <meta>, <link>, <script>, <style> are normally kept under head element.

❖ The <body> element is where you place the **actual content** of your HTML web page. For example, <div>, <h1>, <p>, , <a> are normally kept under body element.

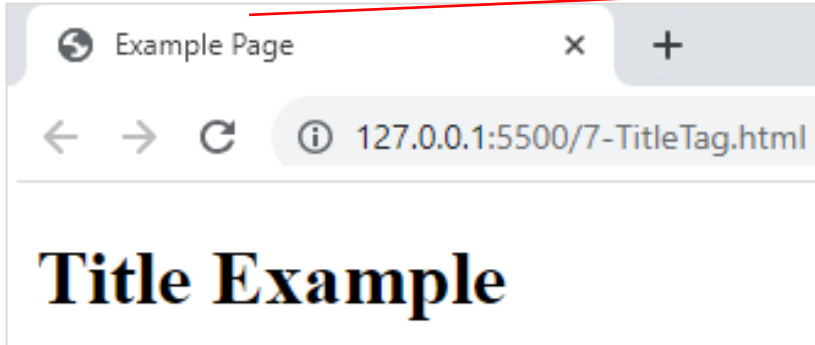
❖ Head will load before the body, therefore, if you are manipulating HTML elements in your JS functions, then place the JS link at the end of the body section because until then all the elements will be loaded. Else, place it inside the head tag.

```
<!DOCTYPE html>
<html>
<head>
  <title>Sample Page</title>
  <meta charset="UTF-8">
  <link rel="stylesheet" href="styles.css">
  <script src="script.js"></script>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is a sample webpage.</p>
  
</body>
</html>
```

Q. What is **Title Tag** in HTML? What are the **4 advantages** of Title tag? **V. IMP.**



- ❖ The <title> tag in HTML is used to define the **title** of a web page.



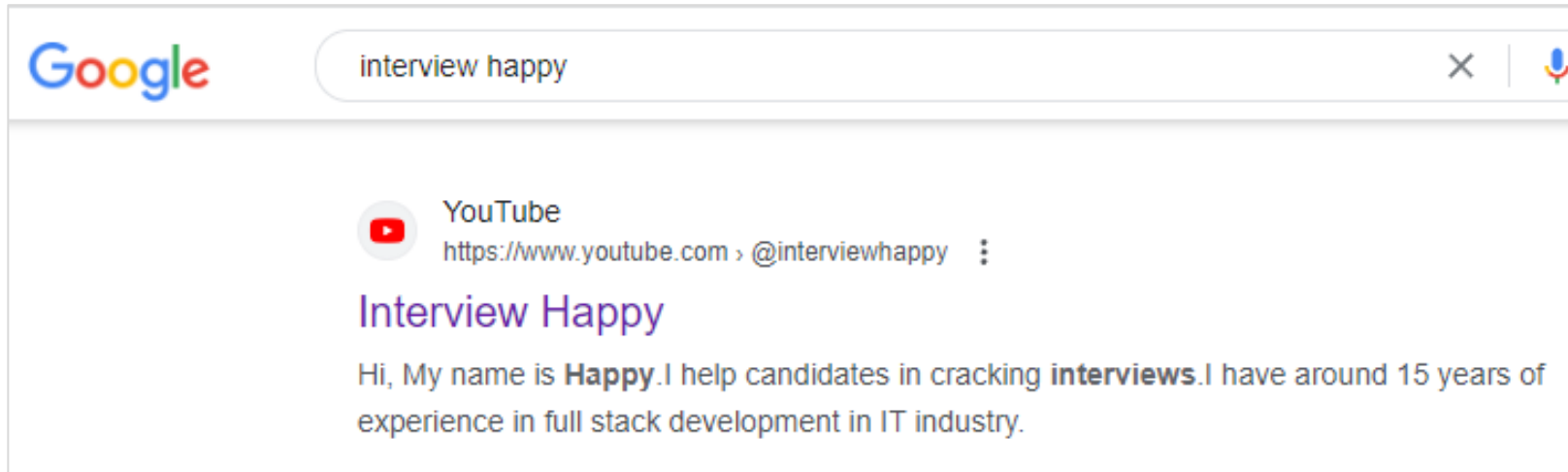
```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>Example Page</title>
  </head>
  <body>
    <h1>Title Example</h1>
  </body>
</html>
```

Q. What is **Title Tag** in HTML? What are the **4 advantages** of Title tag? **V. IMP.**



❖ Advantages of title tag:

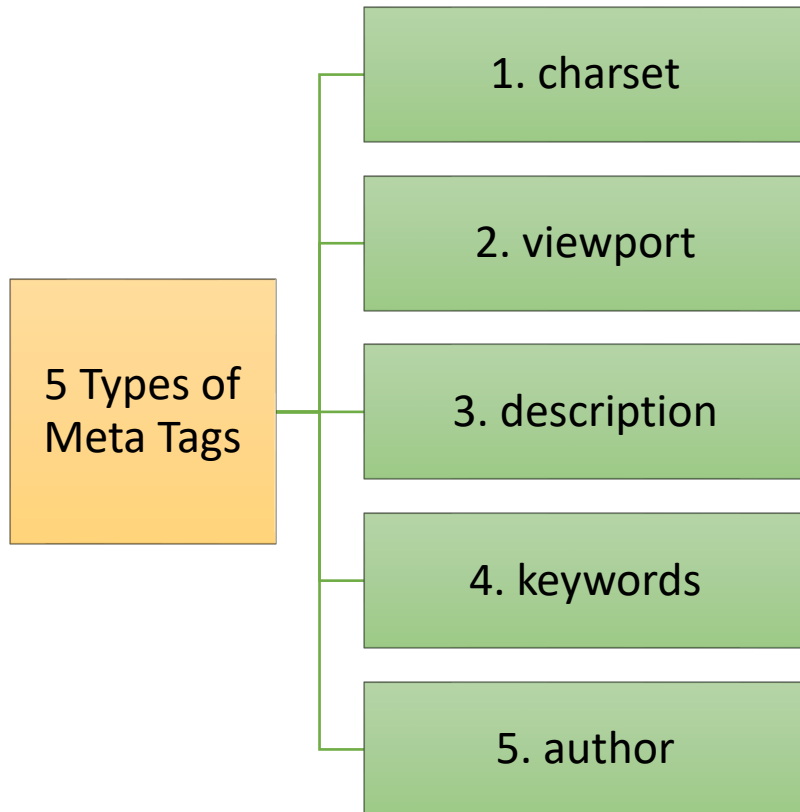
1. **Window Title:** <title> element text is displayed in the tab of the browser.
2. **SEO:** Search engines use the title as the main heading for search results.
3. **Bookmarks/Favorite:** Title tag is used as the default name for the bookmark.
4. **Social Media Sharing:** Title tag is used as the default title in the shared post.



Q. What are **Meta Tags**? What are the **5 types** of meta tags?



- ❖ Meta tags in HTML are elements used to provide metadata or additional information about a web page.



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>Example Page</title>
  </head>
  <body>
    <h1>Title Example</h1>
  </body>
</html>
```

Q. What are **Meta Tags**? What are the **5 types** of meta tags?



1. Character Encoding:

This helps browsers to interpret the different characters in the document. The `charset="UTF-8"` attribute ensures that characters from **different languages** will be displayed correctly in the browser.

2. Responsive Design:

viewport meta tag is crucial for creating mobile-friendly, **responsive web designs** for various screen sizes.

3. Description for SEO:

This meta tag provides description of the content of the page. Search engines may use this for search results.

4. Keywords for SEO:

This meta tag used to be important for search engines, but it's now of less importance.

5. Author Info:

This meta tag can be used to specify the author of the document.

```
<html>
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Interview Happy</title>
    <meta name="description" content="Technical interview preparation" />
    <meta name="keywords" content=".NET, Angular, React, JS, HTML" />
    <meta name="author" content="Happy Rawat" />
  </head>
```

2. HTML Elements & Tags – Part I

Q. What are **HTML Elements**? What is the difference between Element & Tag?

Q. What are the roles and uses of the **<div>** element in HTML?

Q. What is the difference between **<div>** and **** element?

Q. What is the role of **<p>**, **<a>**, **
, **<hr>, ****, ****, **<input>** & **<button>** elements?

Q. What is the role of **header**, **main**, **section**, **footer** & **address** elements in HTML?

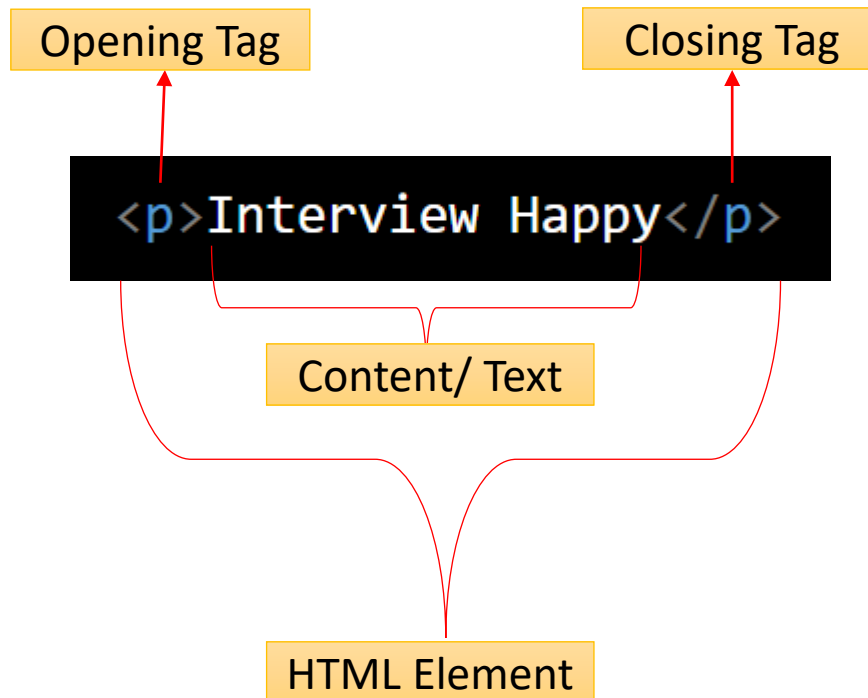
Q. What is the role of **Heading tags** in HTML? How does it impact SEO?

Q. What is the difference between **<section>** & **<article>** elements?

Q. What are **HTML elements**? What is the difference between **Element** & **Tag**? **V. IMP.**



- ❖ HTML elements are the **building blocks** of web pages.
- ❖ Element consists of a start tag, content, and an end tag.
- ❖ Tag is a specific part of an element that denotes the beginning or end of that element.



```
<!DOCTYPE html>
<html>
  <head>
    <title>Title Element</title>
  </head>
  <body>
    <h1>Heading Element</h1>
    <ul><!-- List Element -->
      <li>List Item 1</li>
      <li>List Item 2</li>
    </ul>
    <p>Paragraph Element</p>
    <div>Division Element</div>
    
    <a href="~/AnchorElement.html"></a>
  </body>
</html>
```

Q. What are **HTML elements**? What is the difference between **Element** & **Tag**? **V. IMP.**



Top 10 most common used HTML elements

- 1. Paragraph Element:** Defines a block of text.<p>.
- 2. Heading Elements:** <h1> <h2><h3><h4><h5><h6>.
- 3. Header and Footer Elements:** <header> and <footer>.
- 4. Sectioning Elements:** <section>, <article>, <nav>, <aside>, <main>.
- 5. Image Element:** Embeds images into a webpage. (self-closing).
- 6. List Elements:** (unordered list), (ordered list)), (list item).
- 7. Anchor Element (Link):** Creates hyperlinks to other pages or resources.<a>.
- 8. Division Element:** <div> groups together content for applying styles or scripting.<div>.
- 9. Table Elements:** <table>, <tr> (table row), <th> (table header) and <td> (table data).
- 10. Form Elements:** Used for creating forms.<form>, <input>, <button>, <select>, <textarea>, etc.

Q. What are the roles and uses of the `<div>` element in HTML? **V. IMP.**



❖ The `<div>`(division) element in HTML is a **container** that is used to **group and structure** the content on a webpage.

❖ **Top 3 uses of `<div>` element:**

1. Grouping & structuring Content:

- It allows you to group together related elements.

2. Styling and Layout:

- `<div>` elements are used to apply common styles or css to grouped elements.

3. Scripting:

- JavaScript and other scripting languages can target `<div>` elements to manipulate their content or behavior.

```
<body>
  <div class="container">
    <h1>Heading 1</h1>
    <p>Paragraph 1</p>
  </div>
</body>
```

Heading 1

Paragraph 1



Q. What is the difference between `<div>` and `<span>` element?

- ❖ The `<div>`(division) element in HTML is a **container** that is used to **group and structure** the content on a webpage.
- ❖ The `<span>` element in HTML is an **inline container** used to apply styles or scripting to a specific section of text or content.
- ❖ `<div>` is a block-level element and `<span>` is an inline element.

```
<body>
  <div>
    <h1>Main Heading</h1>
    <p>Interview <span style="color: blue">Happy</span></p>
  </div>
</body>
```

Q. What is the role of `<p>`, `<a>`, `<br>`, `<hr>`, `<em>`, `<img>`, `<input>` & `<button>` elements?



1. `<p>`

- Defines a paragraph of text.

This is a paragraph.

```
<p>This is a paragraph.</p>
```

2. `<a>`

- Creates a hyperlink. The href attribute specifies the url and content is the text of hyperlink.

[link](https://abc.com).

```
<a href="https://abc.com">link</a>.
```

3. `<br>`

- Inserts a line break.

first line.
second line.

```
<p>first line.<br />second line.</p>
```

4. `<hr>`

- Creates a horizontal line break.

```
<hr />
```

Q. What is the role of `<p>`, `<a>`, `<br>`, `<hr>`, `<em>`, `<img>`, `<input>` & `<button>` elements?



1. `<em>`

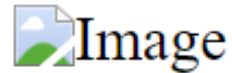
- Emphasizes text displayed in italics.

inline italics text.

```
<p>inline <em>italics text.</em></p>
```

2. `<img>`

- Embeds an image. The `src` attribute specifies the image file, and `alt` provides alternative text.



```
  
<br /><br />
```

3. `<input>`

- Creates an input field within a form.

```
<input type="text" id="user" />  
<br /><br />
```

4. `<button>`

- Creates a clickable button.

Click

```
<button type="button">Click</button>
```



Q. What is the role of **header**, **main**, **section**, **footer** & **address** elements in HTML?

- ❖ `<header>`, `<main>`, `<section>` and `<footer>` are semantic elements, used to define the layout and **structure of a webpage** in a meaningful and organized way.

Website Header

Section 1

Section 2

India

`<header>`

Contains the header content of website.

`<main>`

Contains the main content of the document.

`<footer>`

contains the group related content.

`<address>`

provides contact information.

```
<body>
  <header>
    | <h1>Website Header</h1>
  </header>
  <main>
    | <section id="section1">
      |   <h2>Section 1</h2>
    </section>
    | <section id="section2">
      |   <h2>Section 2</h2>
    </section>
  </main>
  <footer>
    | <address>India</address>
  </footer>
</body>
```

Q. What is the role of **Heading tags** in HTML? How does it impact SEO?



❖ Heading tags in HTML are used to define the **headings of sections** within a webpage.

❖ Top 3 uses of heading tags:

1. **Organization & Readability:** Heading tags organize the content in better way, making it easier for users to understand the organization of the page.
2. **SEO (Search Engine Optimization):** Search engines use headings to understand the importance of content.
3. **Styling and Layout:** For example, same CSS can be applied to all h1 tags to show their importance. That will help in creating consistent layouts.

Electronics Mobiles ~~Samsung~~ Mobiles Samsung Galaxy S Series Samsung Galaxy S20 Samsung Galaxy S20 Ultra

Electronics

Mobiles

Samsung Mobiles

Samsung Galaxy S Series

Samsung Galaxy S20

Samsung Galaxy S20 Ultra

```
<body>
  <h1>Electronics</h1>

  <h2>Mobiles</h2>

  <h3>Samsung Mobiles</h3>

  <h4>Samsung Galaxy S Series</h4>

  <h5>Samsung Galaxy S20</h5>

  <h6>Samsung Galaxy S20 Ultra</h6>
</body>
```

Q. What is the difference between `<section>` & `<article>` elements?



- ❖ The `<section>` element is a generic container used to **group related content together**.
- ❖ The `<article>` element represents a self-contained or **independent piece of content** with a title and content. For example: a blog post, a news article, a forum post, a comment, etc.
- ❖ Article example: offer



```
<body>
  <section>
    <h2>Mobile Section</h2>
    <p>Samsung, Nokia etc</p>
  </section>
  <section>
    <h2>Books Section</h2>
    <p>Technology, Philosophy etc</p>
  </section>

  <article>
    <h1>Festival Sale</h1>
    <p>10% Discount on all items</p>
    <footer>
      <p>Terms and conditions apply</p>
    </footer>
  </article>
</body>
```

3. HTML Elements & Tags – Part II

Q. What are **Root, Parent, Child & Nested** elements?

Q. What are **Empty Elements**?

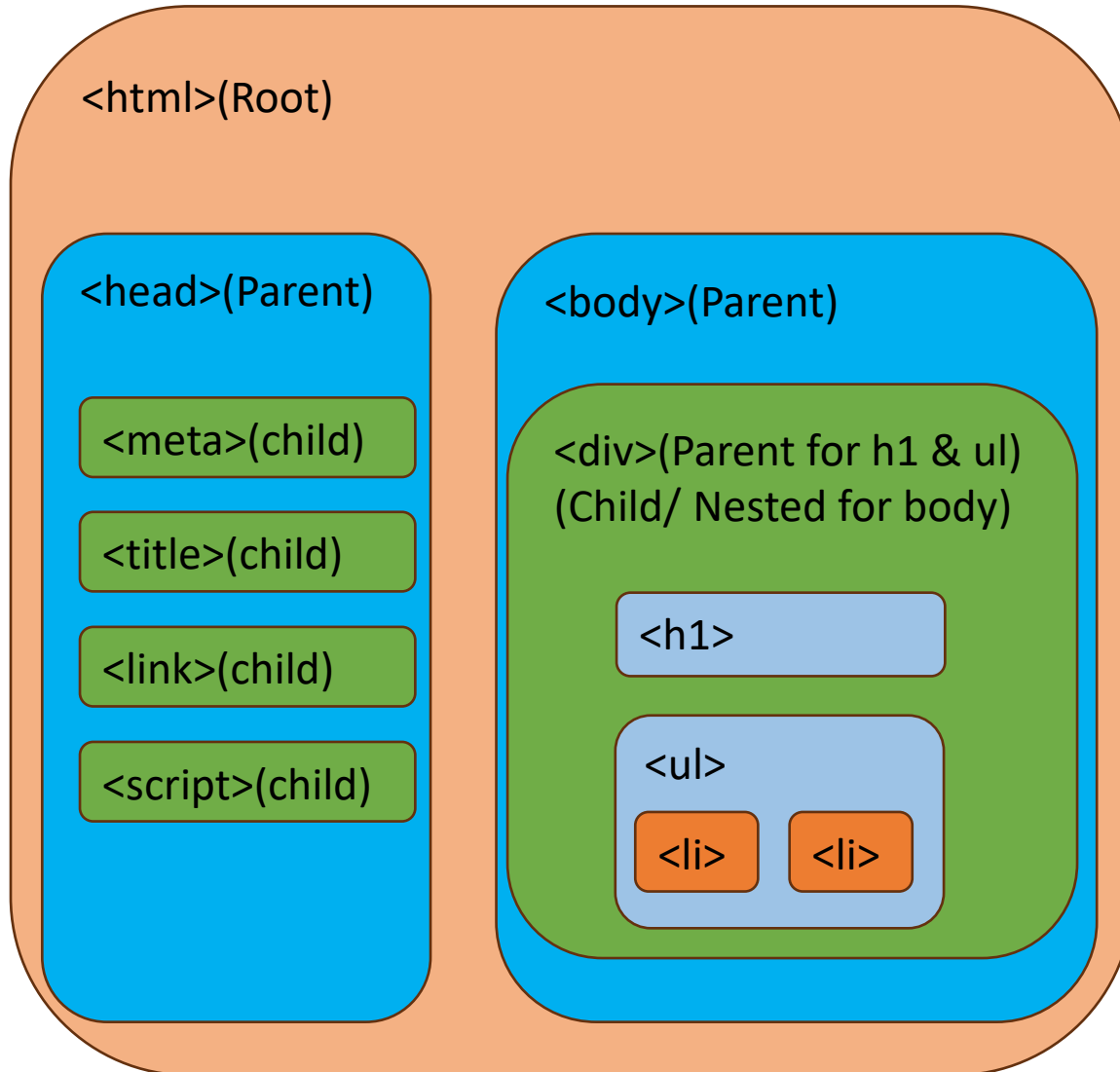
Q. What are **Semantic Elements** in HTML?

Q. Can HTML tags be written in **Uppercase**?

Q. What are the 3 differences between **Block-Level & Inline Elements**?

Q. List all Block-Level & Inline Elements in HTML.

Q. What are **Root**, **Parent**, **Child** & **Nested** elements? **V. IMP.**



```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Nested Elements</title>
    <link rel="stylesheet" href="styles.css" />
    <script src="script.js"></script>
  </head>
  <body>
    <div>
      <h1>Inside container</h1>
      <ul>
        <li>Item 1</li>
        <li>Item 2</li>
      </ul>
    </div>
  </body>
</html>
```

Q. What are **Root, Parent, Child & Nested** elements? **V. IMP.**



1. The **root element** is the highest-level element in the hierarchy of an HTML document. In HTML5, the root element is `<html>`.
2. A **parent element** is an element that contains other child elements.
3. A **child or nested element** is an element that is contained within a parent element. Child and nested elements are same.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Nested Elements</title>
    <link rel="stylesheet" href="styles.css" />
    <script src="script.js"></script>
  </head>
  <body>
    <div>
      <h1>Inside container</h1>
      <ul>
        <li>Item 1</li>
        <li>Item 2</li>
      </ul>
    </div>
  </body>
</html>
```

Q. What are **Empty Elements**?



- ❖ An empty element in HTML is an element that doesn't need content between opening and closing tags.
- ❖ Empty elements are also called a **self-closing** or void elements.
- ❖ Empty elements in HTML: , <input>,
, <hr>, <meta>, <link>, <area>, <base>, <col>, <embed>.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Empty Elements Example</title>
    <link rel="stylesheet" href="style.css" />
    <meta charset="UTF-8" />
  </head>
  <body>
    
    <input type="text" />
    <br />
  </body>
</html>
```

Q. What are **Semantic Elements** in HTML? Is div a semantic element? **V. IMP.**



- ❖ Semantic elements in HTML are elements that provide **meaning to the content** they contain.
- ❖ `<div>` is not a semantic element, because div is a general-purpose structural element. It doesn't give any meaning to the content.

Top 5 Semantic elements

1. `<header>`

2. `<main>`

3. `<section>`

4. `<footer>`

5. `<address>`

```
<body>
  <header>
  |   <h1>Website Header</h1>
  </header>
  <main>
  |   <section id="section1">
  |     <h2>Section 1</h2>
  |   </section>
  </main>
  <aside>
  |   <h2>Aside Content</h2>
  </aside>
  <footer>
  |   <address>India</address>
  </footer>
</body>
```

Q. What are **Semantic Elements** in HTML? Is div a semantic element? **V. IMP.**



❖ 10 more semantic elements:

1. **<progress>** - Displays the progress of a task.
2. **<nav>** - Contains navigation links for a webpage.
3. **<time>** - Represents a specific period in time or a date.
4. **<mark>** - Highlights parts of the text for reference or emphasis.
5. **<summary>** - Provides a summary, caption, or legend for a **<details>** element.
6. **<meter>** - Represents a scalar measurement within a known range (like a gauge).
7. **<figure>** - Contains content like images, illustrations, diagrams, etc., along with a caption.
8. **<details>** - Represents additional information or controls that can be toggled open or closed.
9. **<aside>** - Contains content that is related to the main content but can be considered separate.
10. **<article>** - Represents a self-contained composition in a document, such as a blog post or a news article.

Q. Can HTML tags be written in Uppercase?



- ❖ **Yes**, HTML tags not case-sensitive therefore can be written in uppercase, lowercase, or a combination of both.
- ❖ But it is not recommended as per standards.

```
<!DOCTYPE HTML>
<HTML>
  <HEAD>
    <TITLE>ALL UPPERCASE HTML</TITLE>
  </HEAD>
  <BODY>
    <H1>HELLO, THIS IS AN H1 HEADING</H1>
    <P>THIS IS A PARAGRAPH.</P>
    <A HREF="https://www.abc.com">THIS IS A LINK</A>
  </BODY>
</HTML>
```

Q. What are the 3 differences between **Block-Level** and **Inline Elements**? **V. IMP.**



1. Block-level elements create "blocks" of content.
<div>, <p>, <h1>, , , <table>, <form> etc.
2. By default, Block-level elements start on a new line.
3. You can set both width and height for block-level elements.

```
<!DOCTYPE html>
<html>
  <head>
    <title>BlockLevel Elements</title>
  </head>
  <body>
    <div>block-level element</div>
    <p>block-level element</p>
  </body>
</html>
```

1. Inline elements length depends on their content length.
, <a>, , , , <input>,
 etc.
2. Inline elements do not start on a new line.
3. You can't set width and height for inline elements.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Inline Elements</title>
  </head>
  <body>
    <p>My name is<strong> Happy</strong></p>
    <p>Click <a href="https://abc.com">here</a></p>
  </body>
</html>
```

Q. List all **Block-Level** & **Inline** elements in HTML.



❖ Top 20 block-level elements in HTML:

1. **<h1>, <h2>, <h3>, <h4>, <h5>, <h6>**: Heading elements from highest (h1) to lowest (h6) level.
2. **<header>**: Represents the introductory content or a group of navigational links.
3. **<div>**: The generic container element used for grouping and applying styles.
4. **<footer>**: Represents the footer of a section or document.
5. **<nav>**: Defines a section with navigation links.
6. **<section>**: Defines a section of a document.
7. ****: Creates an ordered (numbered) list.
8. **<p>**: Represents a paragraph of text.
9. ****: Represents a list item in a list.
10. ****: Creates an unordered list.

Q. List all **Block-Level** & **Inline** elements in HTML.



❖ **Top 20 block-level elements in HTML(Continued...):**

- 11. <article>:** Represents a self-contained piece of content, such as a blog post or article.
- 12. <aside>:** Represents content related to the main content but is considered separate.
- 13. <blockquote>:** Represents a section that is quoted from another source.
- 14. <figure>:** Used for encapsulating media and its caption.
- 15. <form>:** Used for creating forms to collect user input.
- 16. <hr>:** Creates a horizontal rule (a thematic break).
- 17. <figcaption>:** Represents a caption for a <figure>.
- 18. <table>:** Defines structure of a table.
- 19. <tr>:** Represents a table row.
- 20. <td>:** Represents a table cell.

Q. List all **Block-Level** & **Inline** elements in HTML.



❖ **Top 20 inline elements in HTML:**

1. ****: A generic container that doesn't have any specific semantic meaning.
2. ****: Indicates strong importance, often displayed as bold text.
3. ****: A generic container used for applying styles or scripting.
4. ****: Emphasizes on text, often displayed as italicized text.
5. **<small>**: Reduces the text size, often used for fine print.
6. **<a>**: Creates hyperlinks to other pages or resources.
7. **<label>**: Provides a label for an input element.
8. **<input>**: Used to create form input fields.
9. **<code>**: Represents a snippet of code.
10. **<cite>**: Specifies the title of a work.

Q. List all **Block-Level** & **Inline** elements in HTML.



❖ **Top 20 inline elements in HTML(Continued...):**

- 11.
:** Inserts a line break, forcing content after it onto a new line.
- 12. <select>:** Creates a dropdown menu for selecting options.
- 13. <time>:** Represents a specific period in time or a date.
- 14. <small>:** Reduces the text size, used for fine print.
- 15. <abbr>:** Defines an abbreviation or acronym.
- 16. :** Indicates deleted or removed text.
- 17. <q>:** Defines a short inline quotation.
- 18. <sup>:** Renders text as superscript.
- 19. <sub>:** Renders text as subscript.
- 20. :** Embeds an image.

4. HTML Attributes

Q. What are **HTML Attributes**? What are the **Types** of HTML attributes?

Q. What are the **Id, Style & Class** attributes of an element? When to use what?

Q. What will happen if two elements have same ids?

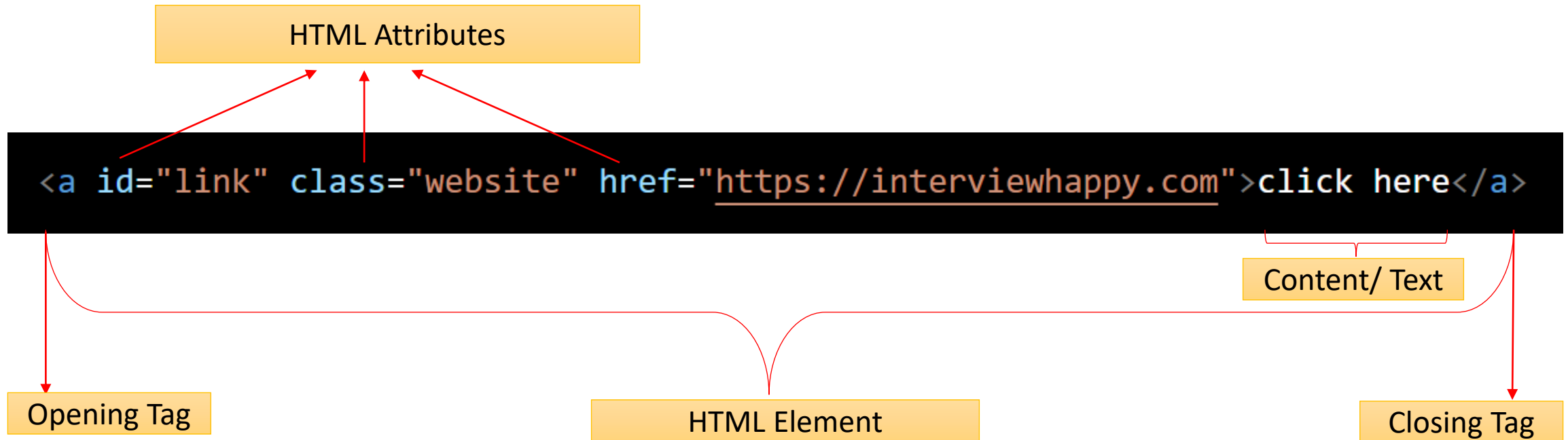
Q. How to specify **Multiple Classes** for single element? What is the style precedence?

Q. What are **Data Attributes** in HTML?

Q. What are **HTML Attributes**? What are the **Types** of HTML attributes? **V. IMP.**



- ❖ HTML attributes provide **additional information** about HTML elements.



Q. What are **HTML Attributes**? What are the **Types** of HTML attributes? **V. IMP.**



❖ **Types of attributes:**

1. Common Attributes (Global Attributes)

- Example: class, id, style, data-*
- They are common and applicable for all the elements.

2. Specific Attributes

- Example: href, src, alt, width, height, target, rel, type, value, name, placeholder, disabled, readonly, checked, selected
- These are specific to the elements.

Q. What are the **Id**, **Style** & **Class** attributes of an element? When to **use** what? **V. IMP.**



- ❖ **id** attribute is used to **uniquely identify** an element on a page.
 - The primary purpose of the id attribute is to allow JavaScript and CSS to target and manipulate specific elements.
- ❖ **style** attribute allows you to apply **inline styles** directly to an individual element.
- ❖ **class** attribute is used to group together multiple elements that **share common styles**.
 - classes are recommended for large website not inline styles.

```
<html>
  <head>
    <title>Id, Style and Class</title>
    <style>
      .highlighted {
        color: ■ red;
      }
    </style>
  </head>
  <body>
    <h1 id="uniqueId">Main heading</h1>

    <p style="color: ■ blue">Para 1</p>

    <p class="highlighted">Para 2</p>
    <p class="highlighted">Para 3</p>
  </body>
</html>
```

Q. What will happen if two elements have **same ids**?



- ❖ May be in browser there is no direct impact, but it will be considered as **invalid HTML**.
- ❖ Invalid HTML means it can lead to **unexpected behavior** in your webpage. For example, problems can occur in css styling and JavaScript interactions.

```
<html lang="en">
  <head>
    <title>uplicate IDs</title>
  </head>
  <body>
    <div id="uniqueId1">First Element</div>
    <p id="uniqueId1">Second Element</p>
  </body>
</html>
```

Q. How to specify **Multiple Classes** for single element? What is the **style precedence**?



- ❖ To specify multiple classes for an element in HTML, you can simply add a **space-separated list of class names** within the class attribute.
- ❖ In case of multiple classes with same style attribute, style defined in the **last class** will take precedence.

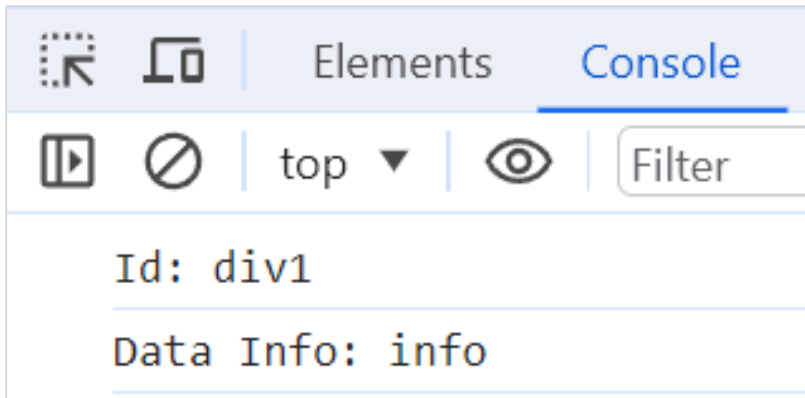
single class
multiple classes

```
<html>
  <head>
    <title>Multiple Classes Example</title>
    <style>
      .class1 {
        color: red;
      }
      .class2 {
        font-size: 20px;
        color: green;
      }
    </style>
  </head>
  <body>
    <div class="class1">single class</div>
    <div class="class1 class2">multiple classes</div>
  </body>
</html>
```

Q. What are **Data Attributes** in HTML? **V. IMP.**



- ❖ Data attributes in HTML provide a way to add **custom data attributes** to add additional informational in elements.
- ❖ Data attributes are specified using the **data-** prefix.
- ❖ Data attributes can be accessed by dataset property in JS.



```
<body>
  <div id="div1" data-info="info">Data Attributes</div>

  <script>
    const element = document.querySelector("div");
    const id = element.id;
    const info = element.dataset.info;
    const dataid = element.dataset.id;

    console.log(`Id: ${id}`);
    console.log(`Data Info: ${info}`);
  </script>
</body>
```

5. HTML Links & Navigation

Q. What are the 5 Types of Links in HTML?

Q. What is the difference between Absolute & Relative URLs?

Q. What is the purpose of the <nav> element in HTML?

Q. What is a Fragment Identifier in a URL?

Q. What is the purpose of the <base> element in HTML.?

Q. How do you add an external stylesheet in your HTML?

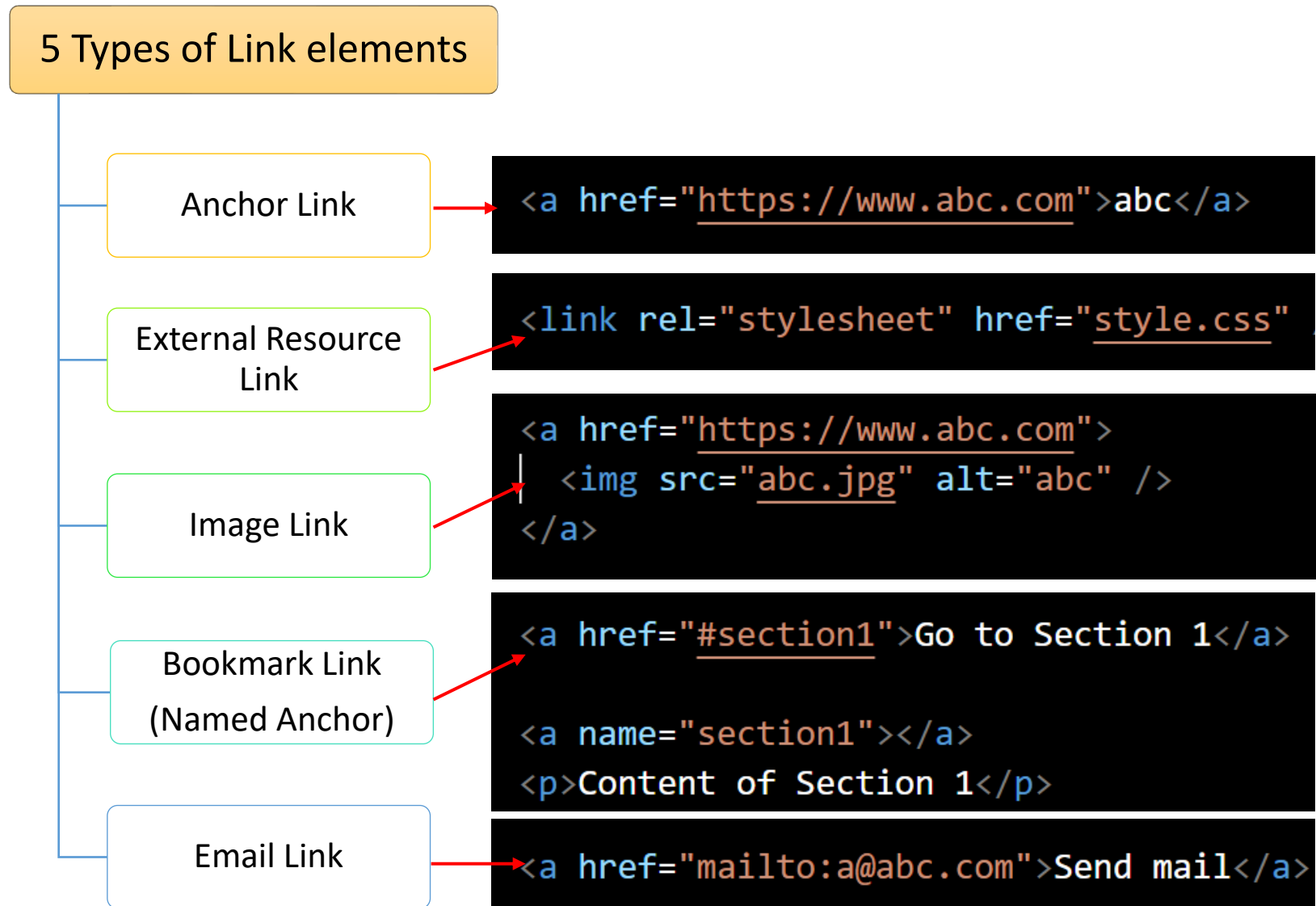
Q. How do you open a link in a new tab?

Q. How do you create an Email Link?

Q. What are the 5 Types of Links in HTML? V. IMP.



1. Anchor Link (<a>) used for navigating from one webpage to another.
2. External Resource Link (<link>) used in the <head> section to connect to an external resource like a stylesheet or an icon.
3. Image Link () used to create a clickable image that leads to another webpage.
4. Bookmark Link (Named Anchor) points to a specific location within a webpage using a named anchor.
5. Email Link creates a clickable link that opens the user's default email client with a pre-filled email address.



Q. What is the difference between **Absolute** and **Relative** URLs? **V. IMP.**



❖ Absolute URLs:

1. Absolute URLs provide the **complete web address** of a resource.
2. Absolute URLs are typically used to link to resources on **different websites**.

```
<h2>Absolute URLs</h2>
<ul>
  <li><a href="http://www.abc.com">HTTP URL</a></li>
  <li><a href="ftp://ftp.abc.com/doc.pdf">FTP URL</a></li>
  <li><a href="mailto:info@abc.com">Mailto URL</a></li>
</ul>
```

❖ Relative URLs:

1. Relative URLs specify the location of a resource in **relation** to the current document. Full url is not required.
2. They are used when linking to resources within the **same website**.

```
<!-- example/index.html -->
<h2>Relative URLs</h2>
<ul>
  <li><a href="page.html">Same Directory</a></li>
  <li><a href="sub/page.html">Subdirectory</a></li>
  <li><a href="../page.html">Parent Directory</a></li>
  <li></li>
  <li><link rel="stylesheet" href="styles/styles.css"/></li>
  <li><script src="scripts/script.js"></script></li>
</ul>
```



Q. What is the purpose of the `<nav>` element in HTML?

- ❖ `<nav>` element in HTML is used to define a section of a web page that contains **navigation links**.
- ❖ `<nav>` element provide clear navigation structure and keep navigation separate from other body content which is good from SEO perspective.

```
<body>
  <header>
    <h1>Website Header</h1>
  </header>
  <nav>
    <ul>
      <li><a href="home.html">Home</a></li>
      <li><a href="about.html">About</a></li>
      <li><a href="services.html">Services</a></li>
      <li><a href="contact.html">Contact</a></li>
    </ul>
  </nav>
</body>
```

- [Home](#)
- [About](#)
- [Services](#)
- [Contact](#)

Q. What is a **Fragment Identifier** in a URL? **V. IMP.**



- ❖ A fragment identifier is used to navigate to a **specific section** of the same webpage.
- ❖ Fragment identifier is preceded by a **# (hash)** symbol.

[Go to Section 1](#)

[Go to Section 2](#)

Section 1

Content for Section 1.

Section 2

```
<body>
  <a href="#section1">Go to Section 1</a>
  <a href="#section2">Go to Section 2</a>

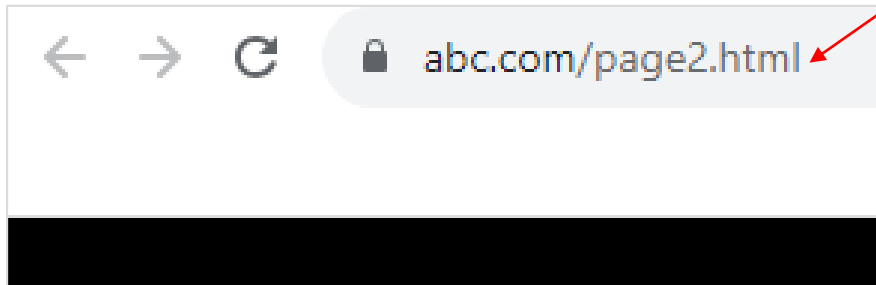
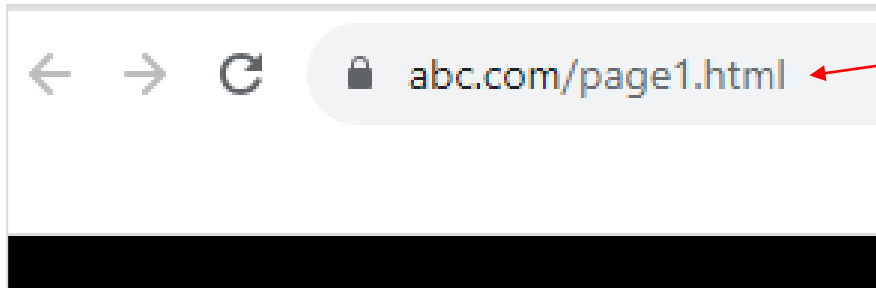
  <section id="section1">
    <h2>Section 1</h2>
    <p>Content for Section 1.</p>
    <div style="height: 500px">Test</div>
  </section>

  <section id="section2">
    <h2>Section 2</h2>
    <p>Content for Section 2.</p>
  </section>
</body>
```



Q. What is the purpose of the `<base>` element in HTML.?

- ❖ The `<base>` element in HTML is used to specify a **base URL** for relative URLs within a document.
- ❖ The `<base>` element is typically placed within the `<head>` section.



```
<html lang="en">
  <head>
    <title>Base Element Example</title>
    <base href="https://www.abc.com/" />
  </head>
  <body>
    <a href="page1.html">Link to Page 1</a>
    <a href="page2.html">Link to Page 2</a>
  </body>
</html>
```

Q. How do you add an **external stylesheet** in your HTML?



- ❖ **<link> element** is an empty element used to link **external resources** like stylesheets to the HTML document.
- ❖ **rel** attribute specifies the **relationship** between the current document and the linked resource.
- ❖ **href** attribute specifies the **location (URL)** of the external stylesheet.

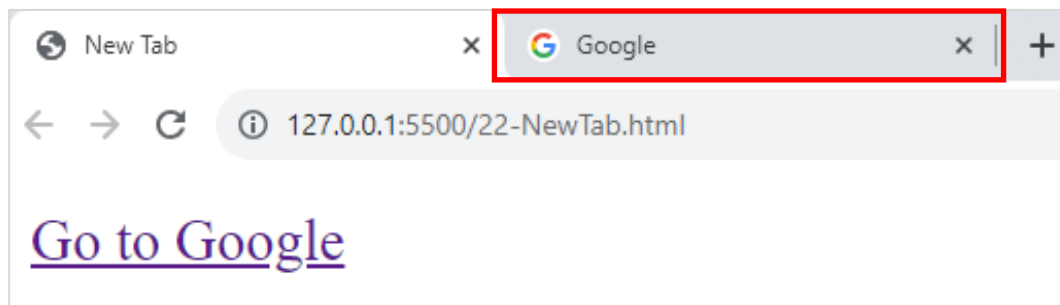
```
<html>
| <head>
| | <link rel="stylesheet" href="styles.css" />
| </head>
| <body>
| | <!-- Body content -->
| </body>
</html>
```

Q. How do you open a link in a new tab?



- ❖ Target attribute specifies how the linked content should be displayed.
- ❖ Target values: `_blank`(new tab), `_self`(default), `_parent`, `_top`

```
<html>
  <head>
    <title>New Tab</title>
  </head>
  <body>
    <a href="https://www.google.com" target="_blank">Go to Google</a>
  </body>
</html>
```

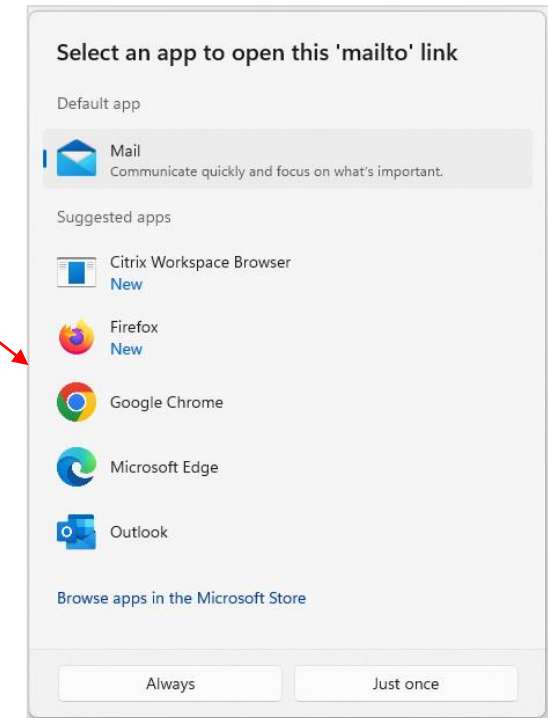
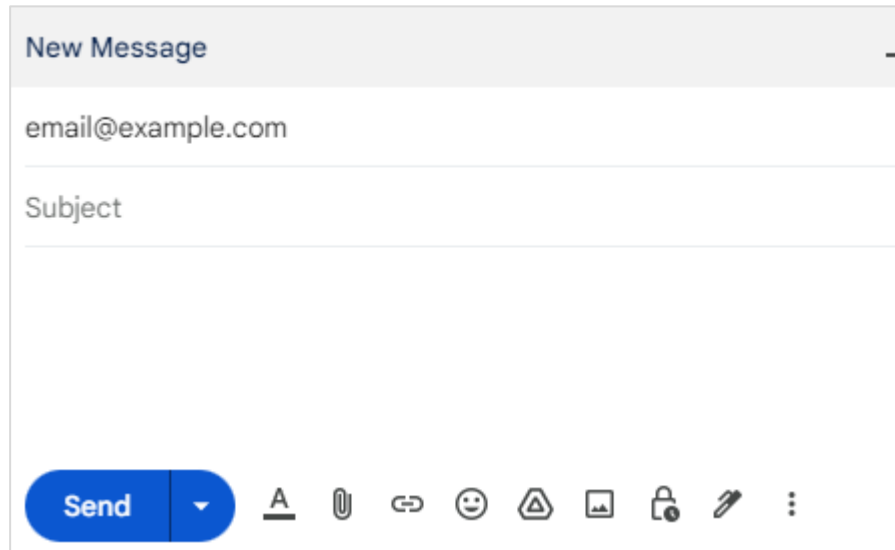
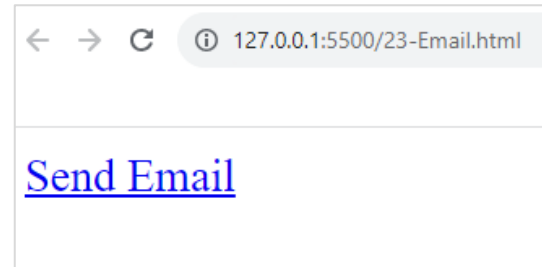


Q. How do you create an **Email Link**?



- ❖ Set the href attribute "mailto:" followed by the email address.
- ❖ This will prompt the user's default email with the specified email address pre-filled in the "To" field. Else it will prompt for the other options.

```
<body>  
  <a href="mailto:email@example.com">Send Email</a>  
</body>
```



6. HTML Lists & Tables

Q. What are the different **Types of Lists** in HTML?

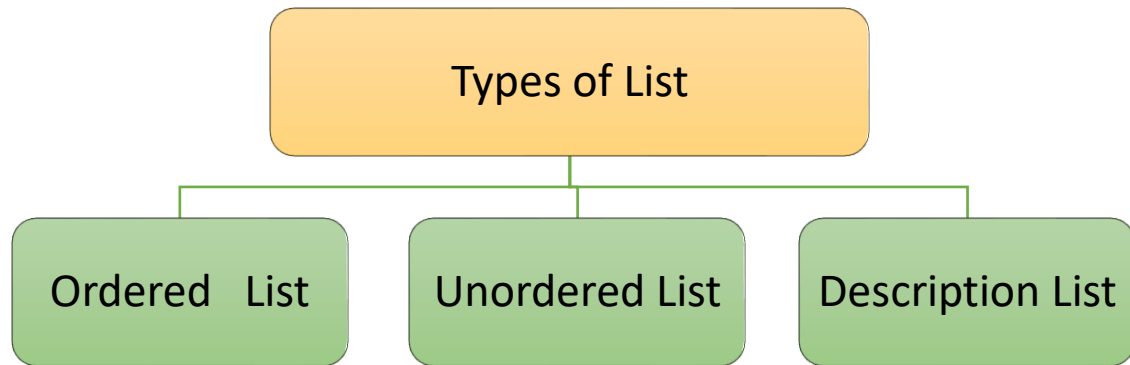
Q. What is a **Nested** List in HTML?

Q. What are **table, tr, th, td** elements? What are table advantages & disadvantages?

Q. What is the **colspan** attribute in HTML?

Q. What is the best way to add a **border** to a table, column and cell?

Q. What are the different **Types of Lists** in HTML? **V. IMP.**



1. An ordered list is a list where the items are **numbered or ordered** in a specific sequence.
2. An unordered list is a list where the items are **marked with bullets**.
3. A description list is used to display a list of terms along with their **corresponding descriptions** or definitions.

Ordered List:

1. Item 1
2. Item 2

```
<h2>Ordered List:</h2>
<ol>
  <li>Item 1</li>
  <li>Item 2</li>
</ol>
```

Unordered List:

- Item A
- Item B

```
<h2>Unordered List:</h2>
<ul>
  <li>Item A</li>
  <li>Item B</li>
</ul>
```

Description List:

Term 1
Description 1

Term 2
Description 2

```
<h2>Description List:</h2>
<dl>
  <dt>Term 1</dt>
  <dd>Description 1</dd>
  <dt>Term 2</dt>
  <dd>Description 2</dd>
</dl>
```

Q. What is a **Nested List** in HTML?



- ❖ A nested list in HTML is a list that is placed within another list item.

Technologies

- Backend
 - Java
 - .NET
 - Node
- Frontend
 - HTML
 - JS
 - React

```
<h1>Technologies</h1>
<ul>
  <li>
    Backend
    <ul>
      <li>Java</li>
      <li>.NET</li>
      <li>Node</li>
    </ul>
  </li>
  <li>
    Frontend
    <ul>
      <li>HTML</li>
      <li>JS</li>
      <li>React</li>
    </ul>
  </li>
</ul>
```



Q. What are **table**, **tr**, **th**, **td elements**? What are table advantages & disadvantages? **V. IMP.**

❖ Table elements:

1. `<table>` is the container for the entire table.
2. `<tr>`(table row) is used to define a row in the table.
3. `<th>`(table header) is used to represent the column headers.
4. `<td>`(table data) is used to represent the regular cells in a table.

Header 1	Header 2
Row 1, Cell 1	Row 1, Cell 2
Row 2, Cell 1	Row 2, Cell 2

Tables are a powerful tool for styling and displaying structured data.

Tables multiple column's structure is not good for mobile devices(not responsive).

```
<body>
  <table border="1">
    <tr>
      <th>Header 1</th>
      <th>Header 2</th>
    </tr>
    <tr>
      <td>Row 1, Cell 1</td>
      <td>Row 1, Cell 2</td>
    </tr>
    <tr>
      <td>Row 2, Cell 1</td>
      <td>Row 2, Cell 2</td>
    </tr>
  </table>
</body>
```



Q. What is the **colspan attribute** in HTML?

- ❖ The colspan attribute is used to **merge multiple cells** horizontally into a single cell.
- ❖ colspan attribute is applicable to **<th>** and **<td>** only.

← → ↻ ⓘ 127.0.0.1:5500/27-Colspan.html

Header 1	Header 2	Header 3
Row 1, Cell 1&2		Row 1, Cell 3
Row 2, Cell 1	Row 2, Cell 2	Row 2, Cell 3

```
<body>
  <table border="1">
    <tr>
      <th>Header 1</th>
      <th>Header 2</th>
      <th>Header 3</th>
    </tr>
    <tr>
      <td colspan="2">Row 1, Cell 1&2</td>
      <td>Row 1, Cell 3</td>
    </tr>
    <tr>
      <td>Row 2, Cell 1</td>
      <td>Row 2, Cell 2</td>
      <td>Row 2, Cell 3</td>
    </tr>
  </table>
</body>
```

Q. What is the best way to **add a border** to a table, column and cell?



- ❖ Setting the common style for table, tr, th and td, then all the multiple tables in your webpage will follow consistent same style and format.

Header 1	Header 2
Row 1, Cell 1	Row 1, Cell 2
Row 2, Cell 1	Row 2, Cell 2

```
<head>
  <title>Styled Table</title>
  <style>
    table {
      border: 2px solid #000;
      width: 25%;
    }

    th,
    td {
      border: 1px solid #000;
      padding: 8px;
      text-align: center;
    }
  </style>
</head>
```

```
<body>
  <table>
    <tr>
      <th>Header 1</th>
      <th>Header 2</th>
    </tr>
    <tr>
      <td>Row 1, Cell 1</td>
      <td>Row 1, Cell 2</td>
    </tr>
    <tr>
      <td>Row 2, Cell 1</td>
      <td>Row 2, Cell 2</td>
    </tr>
  </table>
</body>
```

7. HTML Images & Multimedia

Q. How to embed an **Image** in HTML? Why **alt** attribute is so important?

Q. What are the difference between **JPEG & PNG** image formats?

Q. How to embed **Video** element in HTML? Why we need multiple source elements?

Q. How to embed **Audio** element in HTML?

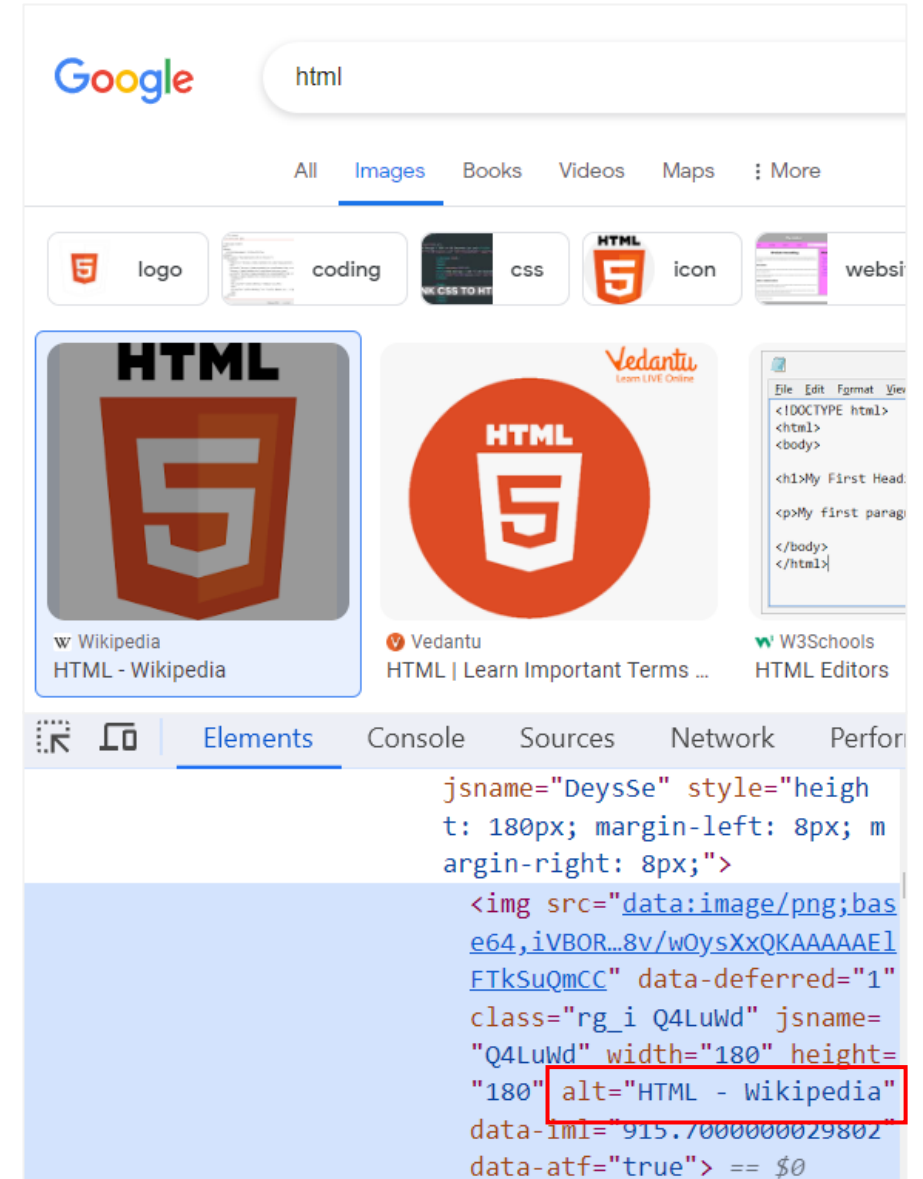
Q. What are the **Autoplay & Loop** attributes in multimedia elements?

Q. How to embed an **Image** in HTML? Why **alt** attribute is so important? **V. IMP.**



- ❖ element is used to embed an image in HTML.
- ❖ The src(soruce) attribute specifies the source file of the image.
- ❖ The alt(alternative text) attribute provides alternative text that describes the content of the image which is important for SEO(for searching).

```
<html lang="en">
  <head>
    <title>Image Example</title>
  </head>
  <body>
    
  </body>
</html>
```



Q. What are the difference between **JPEG** and **PNG** image formats?



JPEG	PNG
1. JPEG stands for Joint Photographic Experts Group.	PNG stands for Portable Network Graphics
2. JPEG uses lossy compression , which means some data is discarded to reduce image size.	PNG uses lossless compression , meaning no data is lost during compression.
3. Due to lossy compression, lower image quality .	Due to lossless compression, higher image quality.
4. Due to lossy compression, Image size is reduced.	Due to lossless compression, the image has a comparatively larger size.

Q. What are the difference between **JPEG** and **PNG** image formats?



JPEG	PNG
1. JPEG stands for Joint Photographic Experts Group.	PNG stands for Portable Network Graphics
2. JPEG uses lossy compression , which means some data is discarded to reduce image size.	PNG uses lossless compression , meaning no data is lost during compression.
3. Due to lossy compression, lower image quality .	Due to lossless compression, higher image quality.
4. Due to lossy compression, Image size is reduced.	Due to lossless compression, the image has a comparatively larger size.

Q. How to embed Video in HTML? Why we need multiple source elements?



1

<video> is the element used to embed videos.

2

controls attribute adds video controls like play, pause & volume.

3

<source> element specifies multiple sources for the video. If one video format is not compatible with browser, then other will execute.

4

src attribute specifies the path to the video file.

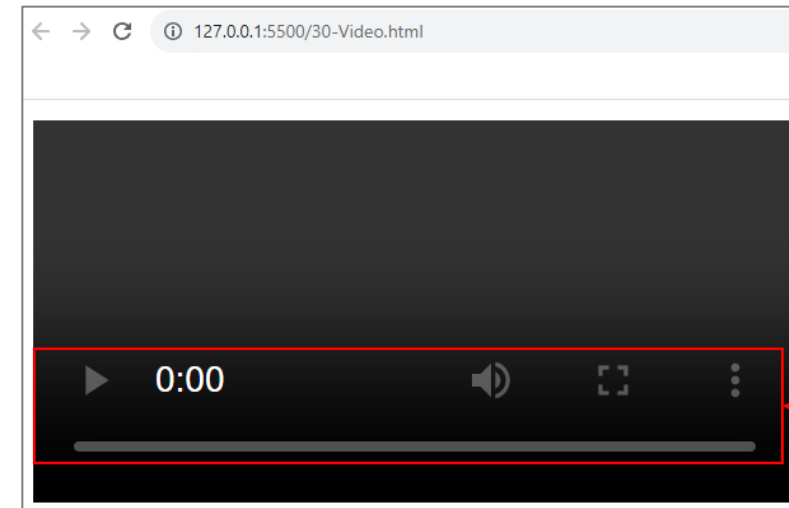
5

type attribute indicates the MIME type.

6

The **text** "Your browser..." is a fallback message for browsers that do not support the **<video>** element.

```
<body>
  <video controls>
    <source src="filepath.mp4" type="video/mp4" />
    <source src="filepath.webm" type="video/webm" />
    Your browser does not support the video tag.
  </video>
</body>
```



Q. How to embed **Audio** in HTML?



1

<audio> is the element used to embed audios.

2

controls attribute adds video controls like play, pause & volume.

3

<source> element specifies multiple sources for the audio. If one audio format is not compatible with browser, then other will execute.

4

src attribute specifies the path to the audio file.

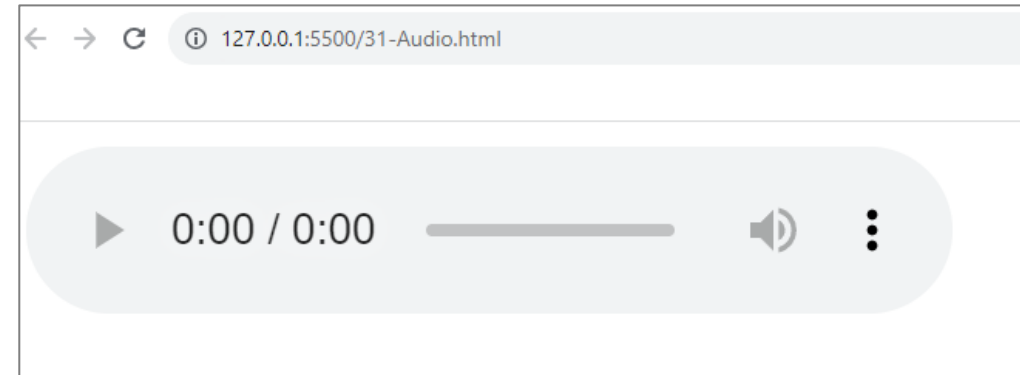
5

type attribute indicates the MIME type.

6

The **text** "Your browser..." is a fallback message for browsers that do not support the **<audio>** element.

```
<body>
  <audio controls>
    <source src="filepath.mp3" type="audio/mp3" />
    <source src="filepath.ogg" type="audio/ogg" />
    Your browser does not support the audio element.
  </audio>
</body>
```



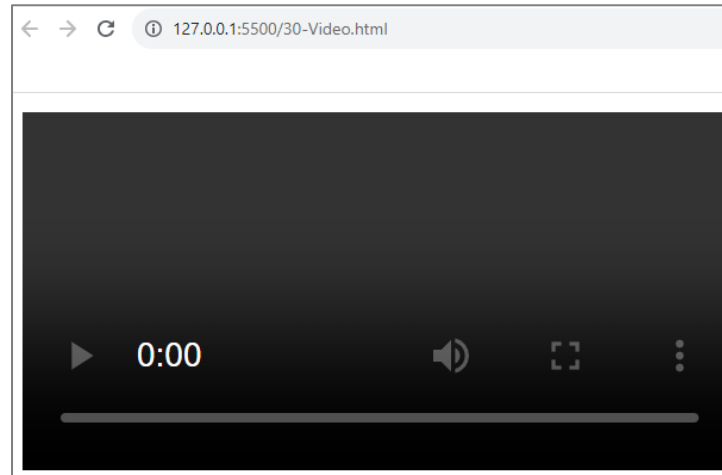
Q. What are the **Autoplay** & **Loop** attributes in multimedia elements?



- ❖ The autoplay attribute allows media content to **start playing automatically** as soon as the page is loaded.
- ❖ The loop attribute, allows the media content to **restart** whenever it reaches the end.

```
<h2>Autoplay</h2>
<video src="video.mp4" autoplay controls>
  Your browser does not support the video element.
</video>
```

```
<h2>Loop</h2>
<video src="video.mp4" loop controls>
  Your browser does not support the video element.
</video>
```



8. HTML Forms

Q. What are **HTML Forms** and what are its **advantages**?

Q. Q. What are HTML **Form Elements** and their **Main Attributes**?

Q. How to **Validate** for empty input field in form?

Q. What is the role of the **<textarea>** element in a form?

Q. What are the **Types of Input elements** in a form?

Q. How to create **radio buttons** and **checkboxes**? How to group them?

Q. What is the purpose of the **<select>** element in a form?

Q. What are **HTML Forms** and what are its **advantages**? **V. IMP.**



- ❖ Forms provide a way to collect and **submit** user data on a webpage.

Advantages of Form element

1. Data collection & organization is simple
2. Data validation is simple
3. Data submission on server is easy
4. Styling and Layout
5. Compatible with all browsers

Login Form

Username:

Password:

Q. What are **HTML Form Elements** and their **Main Attributes**? **V. IMP.**



Form elements & Attributes

1. form element

2. action attribute

3. method attribute

4. label element

5. for attribute

6. input element

7. type attribute

8. name attribute

```
<form action="/submit" method="post">
  <label for="username">Username:</label>
  <input type="text" id="username" name="username" />
  <br /><br />
  <label for="password">Password:</label>
  <input type="password" id="pwd" name="password" />
  <br /><br />
  <input type="submit" value="Submit" />
</form>
```

Login Form

Username:

Password:

Q. What are **HTML Form Elements** and their **Main Attributes**? **V. IMP.**



❖ **Form elements and attributes**

1. **<form>** element is the container of form elements.
2. The **action** attribute specifies the **URL** where the form data will be sent when it is submitted.
3. The **method** attribute defines the **HTTP method** to be used when sending the form data.
4. The **<label>** element is used to provide a label or **description** for an input field. The **for** attribute **associate** the label with a specific input field.
5. The **<input>** element is used to **create controls**.
6. The **type** attribute specifies the type of input field.
7. The **name** attribute is used to **define a name** for the input field.

Q. How to **Validate** for empty input field in form?



❖ **required** attribute is used to make the input field mandatory.

```
<body>
  <h3>Login Form</h3>
  <form action="/submit" method="post">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required />
    <br /><br />
    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required />
    <br /><br />
    <input type="submit" value="Submit" />
  </form>
</body>
```

Login Form

Username:

Password:

Submit



Please fill out this field.



Q. What is the role of the `<textarea>` element in a form?

- ❖ The `<textarea>` element is used in HTML forms to create a **multi-line text** input field.
- ❖ `rows` attribute in `textarea` identify the **number of lines** in the text area.
- ❖ `cols` attribute in `textarea` identify the **number of characters** in one row.

```
<body>
  <form>
    <label for="comments">Comments:</label><br />
    <textarea id="comments" name="comments"
      rows="4" cols="50"></textarea>
    <br />
    <input type="submit" value="Submit" />
  </form>
</body>
```

Comments:

1234567891012345678910123456789101234567891012345678
2
3
4|

Submit

Q. What are the **Types of Input elements** in a form?



Types of Input elements

Text Input

```
<input type="text" id="uid" name="uid" />  
<br /><br /><br />
```

Happy

Password Input

```
<input type="password" id="pwd" name="pwd" />  
<br /><br /><br />
```

.....

File Input

```
<input type="file" id="file" name="file" />  
<br /><br /><br />
```

Choose File No file chosen

Submit Button

```
<input type="submit" value="Submit" />  
<br /><br /><br />
```

Submit

Reset Button

```
<input type="reset" value="Reset" />  
<br /><br /><br />
```

Reset

Checkbox

Radio Button

Select

Q. What are the **Types of Input elements** in a form?



Types of Input elements

Text Input

Password Input

File Input

Submit Button

Reset Button

Checkbox

Radio Button

Select

1. Text Input: Allows users to type alphanumeric characters. Commonly used for names, email addresses, passwords, etc.

2. Password Input: Masks the characters entered by the user, typically used for sensitive information like passwords.

3. File Input: Allows users to upload files.

4. Submit Button: Creates a button that submits the form data to a server for processing.

5. Reset Button: Creates a button that resets all the form fields back to their default values.

Q. How to create **radio buttons** and **checkboxes**? How to **group** them?



- ❖ Radio buttons are used when you want the user to **select exactly one option** from a group of options.

```
<label> <input type="radio" name="gender" value="male" /> Male </label><br />
<label> <input type="radio" name="gender" value="female" /> Female </label><br />
<label> <input type="radio" name="gender" value="other" /> Other </label><br />
```

☒ Male
☐ Female
☐ Other

- ❖ Checkboxes are used when you want the user to be able to **select multiple options** from a list.

```
<label> <input type="checkbox" name="hobby" value="music" /> Music </label><br />
<label> <input type="checkbox" name="hobby" value="sports" /> Sports </label><br />
<label> <input type="checkbox" name="hobby" value="travel" /> Travel </label>
```

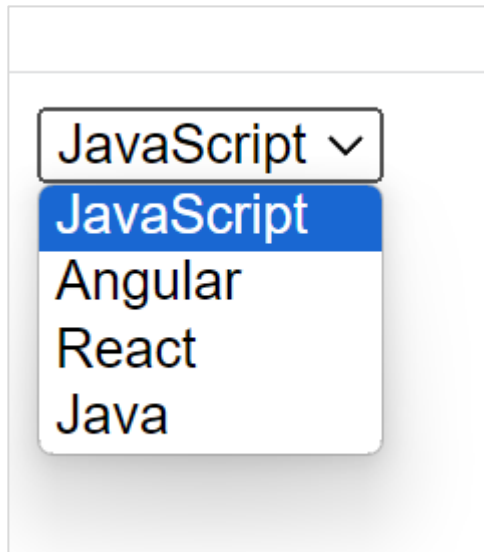
☒ Music
☒ Sports
☐ Travel

- ❖ All radio buttons or checkboxes within a group should share the **same name** attribute.



Q. What is the purpose of the `<select>` element in a form?

- ❖ The `<select>` element in HTML is used to create a **dropdown menu** within a form.
- ❖ Dropdown menus are space-efficient compared to radio-buttons and checkboxes for displaying a **long list of options**.



```
<body>
  <form>
    <select name="Technology">
      <option value="1">JavaScript</option>
      <option value="2">Angular</option>
      <option value="3">React</option>
      <option value="4">Java</option>
    </select>
  </form>
</body>
```

9. Best Practices, Boilerplate & Deployment

Q. What are **Top 10 Best Practices** for writing HTML code?

Q. What is HTML5 **Boilerplate** code?

Q. How do you **optimize** an HTML website for Performance?

Q. What is **SEO**? What are 5 HTML Best Practices for SEO?

Q. What are the 5 steps required for HTML website **deployment**?

Q. What are top 10 Best Practices for writing HTML code? **V. IMP.**



❖ HTML Bad Practices

```
<!-- 1. Missing DOCTYPE declaration -->
<html>
  <head>
    <!-- 2. Missing charset meta tag -->
    <title>Interview Happy</title>
  </head>
  <body>
    <!-- 3. Missing semantic header tag -->
    <div id="Header">
      <h1>Interview Happy</h1>
    </div>

    <!-- 4. Meaningless ID's and classes -->
    <div id="xyz" class="abc">
      <!-- 5. Incorrect capitalization of tags -->
      <P>Help in cracking interviews</P>
      <!-- 6. Missing Alt Text for Images -->
      
    </div>
  </body>
</html>
```

❖ HTML Best Practices

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport"
    content="width=device-width, initial-scale=1.0" />
    <title>Interview Happy</title>
    <link rel="stylesheet" href="styles.css" />
  </head>
  <body>
    <header>
      <h1>Interview Happy</h1>
    </header>
    <main>
      <section id="content">
        <p>Help in cracking interviews</p>
        
      </section>
    </main>
  </body>
</html>
```

Q. What are top 10 Best Practices for writing HTML code? V. IMP.



❖ 7 Best practices for HTML:

1. Proper DocType declaration
2. Proper Meta Tags
3. Use Semantic Elements.
4. Give meaningful ID's and class names.
5. Give Lowercase Element Names.
6. Alternative Text for Images by alt attribute.
7. Use of proper Header Elements.(h1,h2,h3,h4,h5,h6)
8. Use External CSS Link for styling.
9. Proper Indentation and Formatting for readability.
10. Always give title for the website page.

❖ HTML with best practices

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport"
    content="width=device-width, initial-scale=1.0" />
    <title>Interview Happy</title>
    <link rel="stylesheet" href="styles.css" />
  </head>
  <body>
    <header>
      <h1>Interview Happy</h1>
    </header>
    <main>
      <section id="content">
        <p>Help in cracking interviews</p>
        
      </section>
    </main>
  </body>
</html>
```

Q. What is an HTML5 Boilerplate?



- ❖ An HTML5 boilerplate is a **basic template** for building modern websites using HTML5.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>HTML 5 Boilerplate</title>
    <link rel="stylesheet" href="style.css" />
  </head>
  <body>
    <!-- Website Content -->
    <script src="index.js"></script>
  </body>
</html>
```

Q. How do you optimize an HTML website for **Performance**?



❖ 8 Best Practices for performance improvement in HTML:

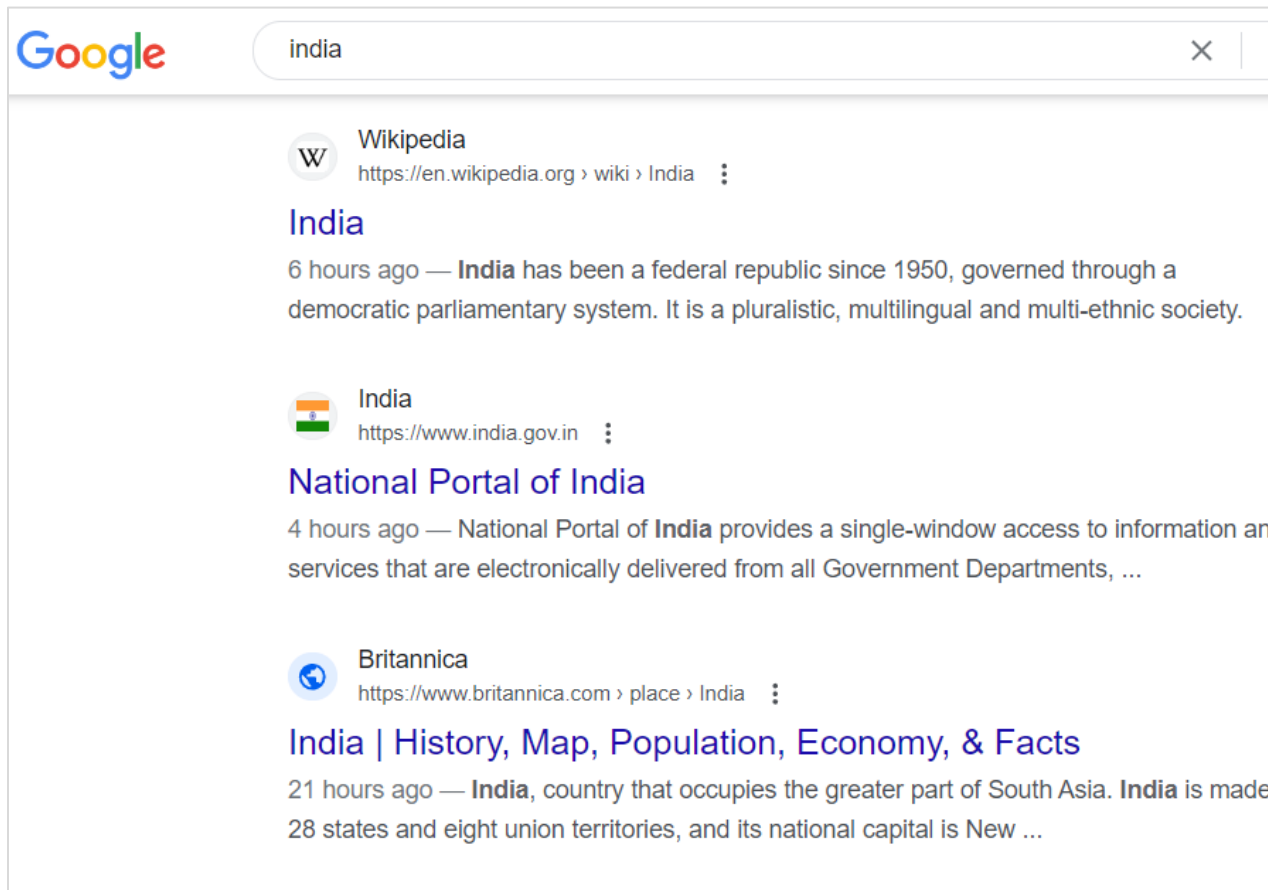
1. **Minimize HTTP Requests:** Reduce the number of HTTP requests by combining CSS and JavaScript files.
2. **Use External CSS and JavaScript Files**
3. **Minify CSS, JavaScript, and HTML:** Remove unnecessary whitespace, comments & formatting to reduce file sizes.
4. **Compress Images:** Use tools to compress images and choose appropriate formats.
5. **Leverage Browser Caching:** Set cache headers to allow resources to be stored locally, reducing the need for repeated requests.
6. **Load Critical Resources First.**
7. **Use Async and Await in JS.**
8. **Implement Lazy Loading** to improve initial load times.



Q. What is **SEO**? What are **5 HTML Best Practices** for SEO?



- ❖ SEO(Search Engine Optimization) is the practice of optimizing a website to improve its **visibility** and **ranking** in search engine results.



5 HTML Best Practices for SEO

1. Use Semantic HTML Elements

2. Optimize Page Titles

3. Use Meta Description Tags

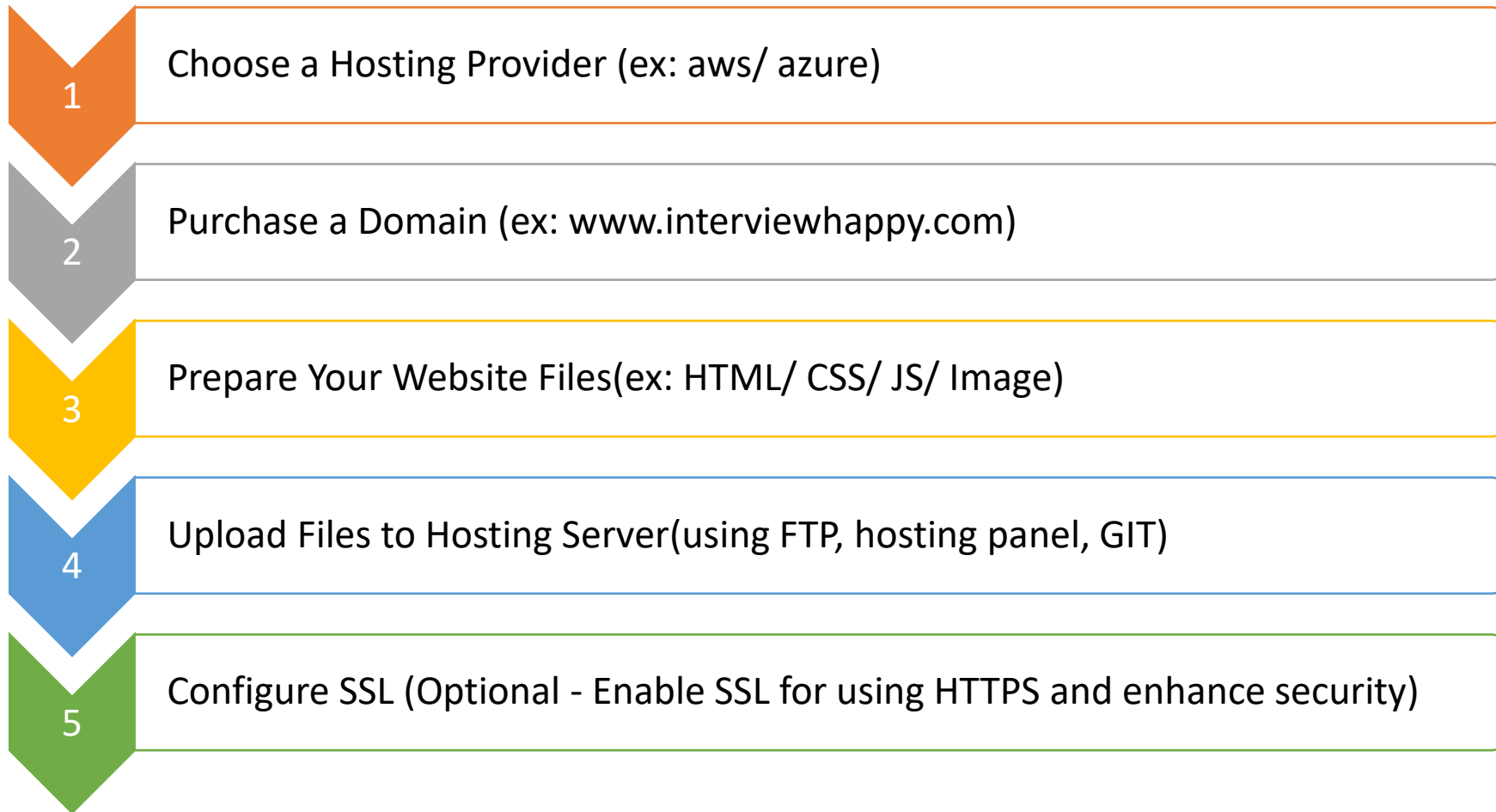
4. Use Proper Heading Tags

5. Optimize Image Alt Attributes

Q. What are the 5 steps required for HTML website deployment?



❖ 5 Basic Steps for HTML website deployment:



10. HTML Responsiveness, Frameworks & Others

Q. What is **Responsive Design**?

Q. What are the top 3 things required to make a website Responsive in HTML?

Q. What are some **Responsive Design Frameworks** for HTML & CSS?

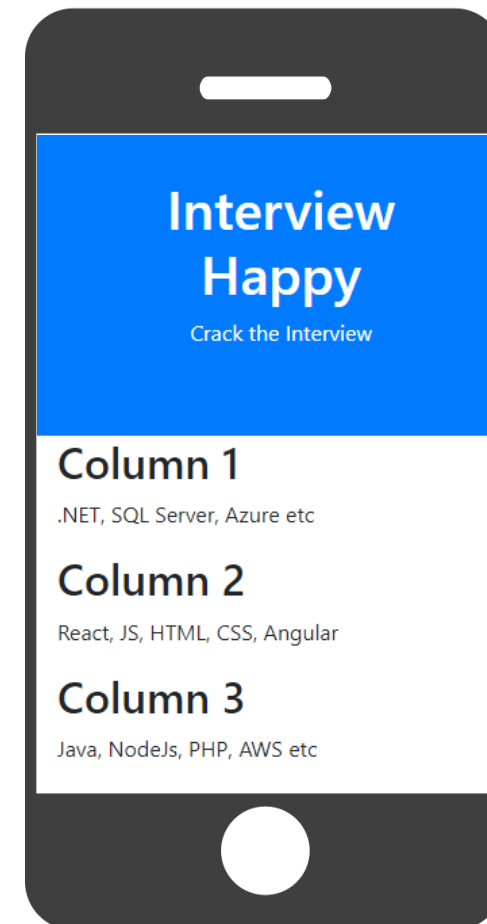
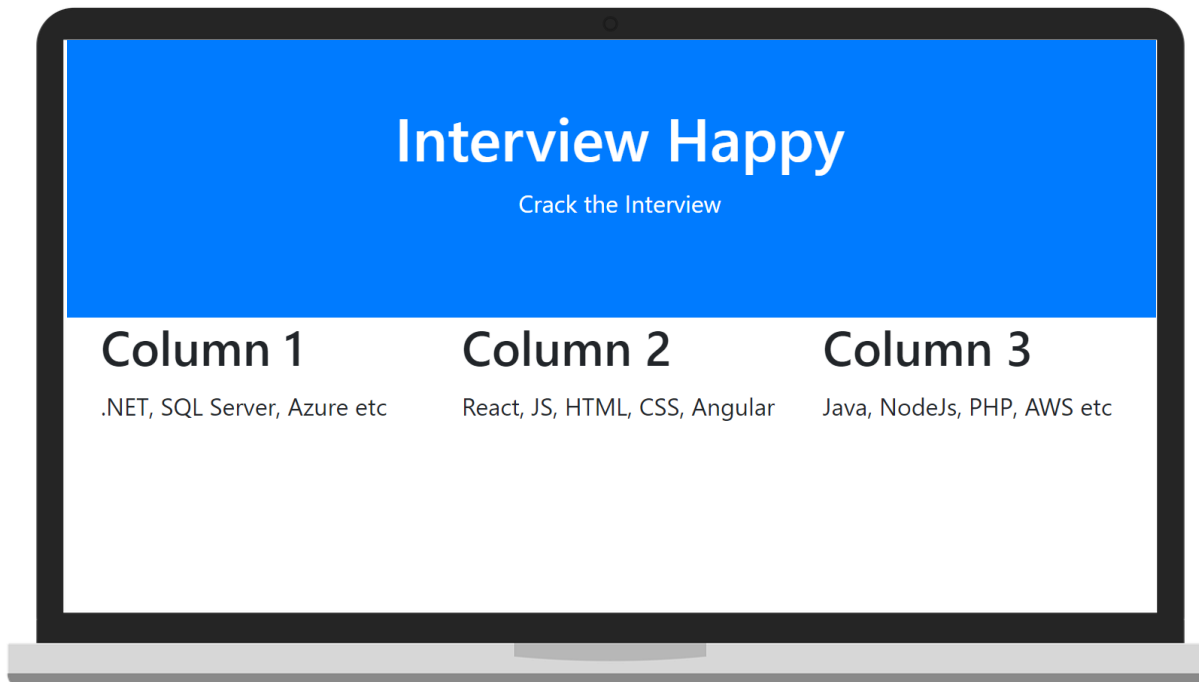
Q. What are **HTML Entities**? How do you display special characters in HTML?

Q. What are the **Top 5 Emerging Trends** in HTML development?

Q. What is **Responsive Design**? **V. IMP.**



- ❖ Responsive design is a practice of creating web pages that **adapt** and display well on various devices and **screen sizes**.



Q. What are the top 3 things required to make a website **Responsive** in HTML?

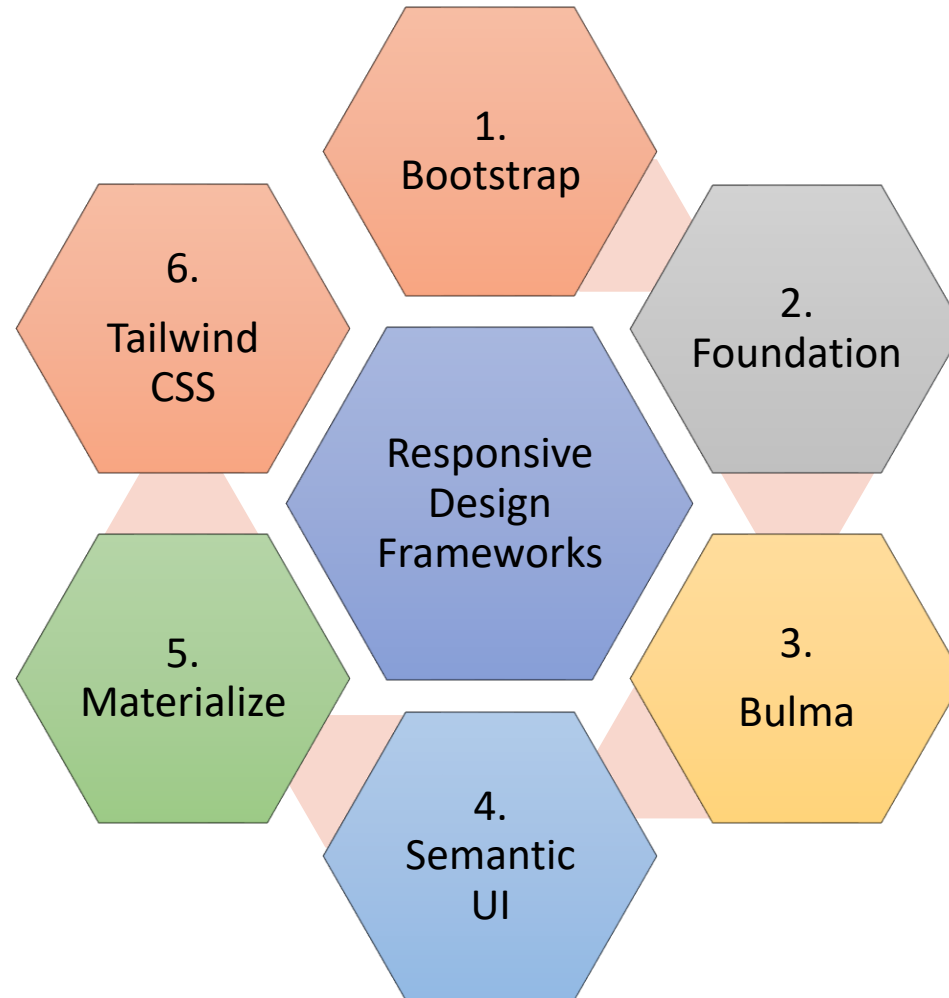


1. **Set the Viewport Meta Tag** : This tag tells the browser to adjust the width of the page to match the screen's width.
2. **Use Relative Units**: Instead of using fixed units like pixels, use relative units like percentages
3. **Media Queries**: They set different sizes for different screen types.

```
.box {  
  width: 30%;  
}  
  
@media (max-width: 768px) {  
  .box {  
    width: 45%;  
  }  
}
```

```
<!DOCTYPE html>  
<html lang="en">  
  <head>  
    <meta charset="UTF-8" />  
    <meta name="viewport"  
      content="width=device-width, initial-scale=1.0" />  
    <title>Responsive HTML Example</title>  
    <link rel="stylesheet" href="39-MakeResponsive.css"  
  </head>  
  <body>  
    <div class="container">  
      <div class="box" width="33%">Box 1</div>  
      <div class="box" width="33%">Box 2</div>  
      <div class="box" width="33%">Box 3</div>  
    </div>  
  </body>  
</html>
```

Q. What are some **Responsive Design Frameworks** for HTML & CSS?



Q. What are **HTML Entities**? How do you display **special characters** in HTML?



- ❖ HTML entities are special codes or sequences of characters used to represent reserved or special characters in HTML.

Top 10 Special Characters in HTML

Ampersand: &

Lesser than: <

Greater than: >

Euro symbol: €

Copyright symbol: ©

Trademark symbol: ™

Non-breaking space:

Double quotation mark: "

Single quotation mark: '

Registered trademark symbol: ®

```
<body>
  <h5>Top 10 Special Characters in HTML</h5>
  <p>Ampersand: &amp;</p>
  <p>Lesser than: &lt;</p>
  <p>Greater than: &gt;</p>
  <p>Euro symbol: &euro;</p>
  <p>Copyright symbol: &copy;</p>
  <p>Trademark symbol: &trade;</p>
  <p>Non-breaking space: &nbsp;</p>
  <p>Double quotation mark: &quot;</p>
  <p>Single quotation mark: &apos;</p>
  <p>Registered trademark symbol: &reg;</p>
</body>
```

Q. What are the **Top 5 Emerging Trends** in HTML development?



1. Web Components:

Web Components are a set of web platform APIs that allow you to create custom, reusable HTML elements.

2. Progressive Web Apps (PWAs)

PWAs are web applications that provide a native app-like experience to users.

3. Single Page Applications (SPAs):

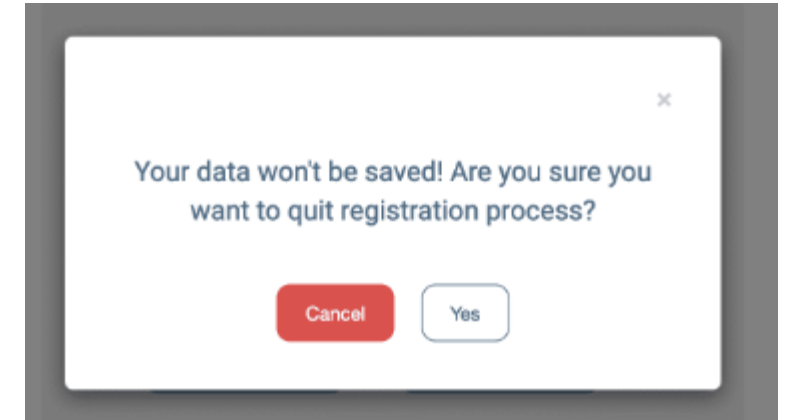
SPAs load a single HTML page and dynamically update content as the user interacts with the app.

4. 3D Graphics and Virtual Reality (VR)

VR refers to the integration of virtual reality experiences within a web browser using HTML, CSS, and JS. Example: WebGL & WebVR.

5. Augmented Reality (AR):

AR combines the physical world with digital information or content. Example: ARCore and ARKit for web development.



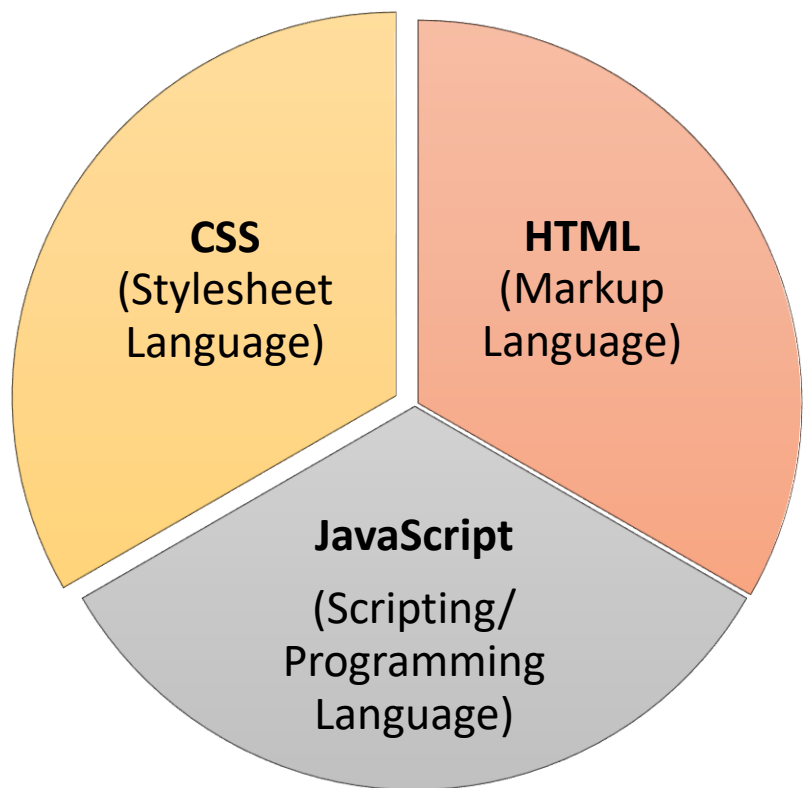
1. CSS Basics & Implementation

- Q. What is CSS? What are the 5 ways to Implement CSS in HTML?
- Q. What is Inline Style in CSS? When to use it in real applications?
- Q. What is Internal Stylesheet in CSS? When to use it in real applications?
- Q. What is External Stylesheet in CSS? When to use it in real applications?
- Q. What are the 5 advantages of external stylesheet?
- Q. How do you Include CSS in a webpage or HTML?
- Q. How to implement CSS using @import rule?
- Q. What is CSS Preprocessors? What is SCSS?
- Q. What are the 3 Types of CSS Preprocessors?

Q. What is **CSS**? What are the 5 ways to **Implement CSS** in HTML? **V. IMP.**



- ❖ CSS(Cascading Style Sheets) is a **stylesheet language** used to control the **presentation** of web pages.



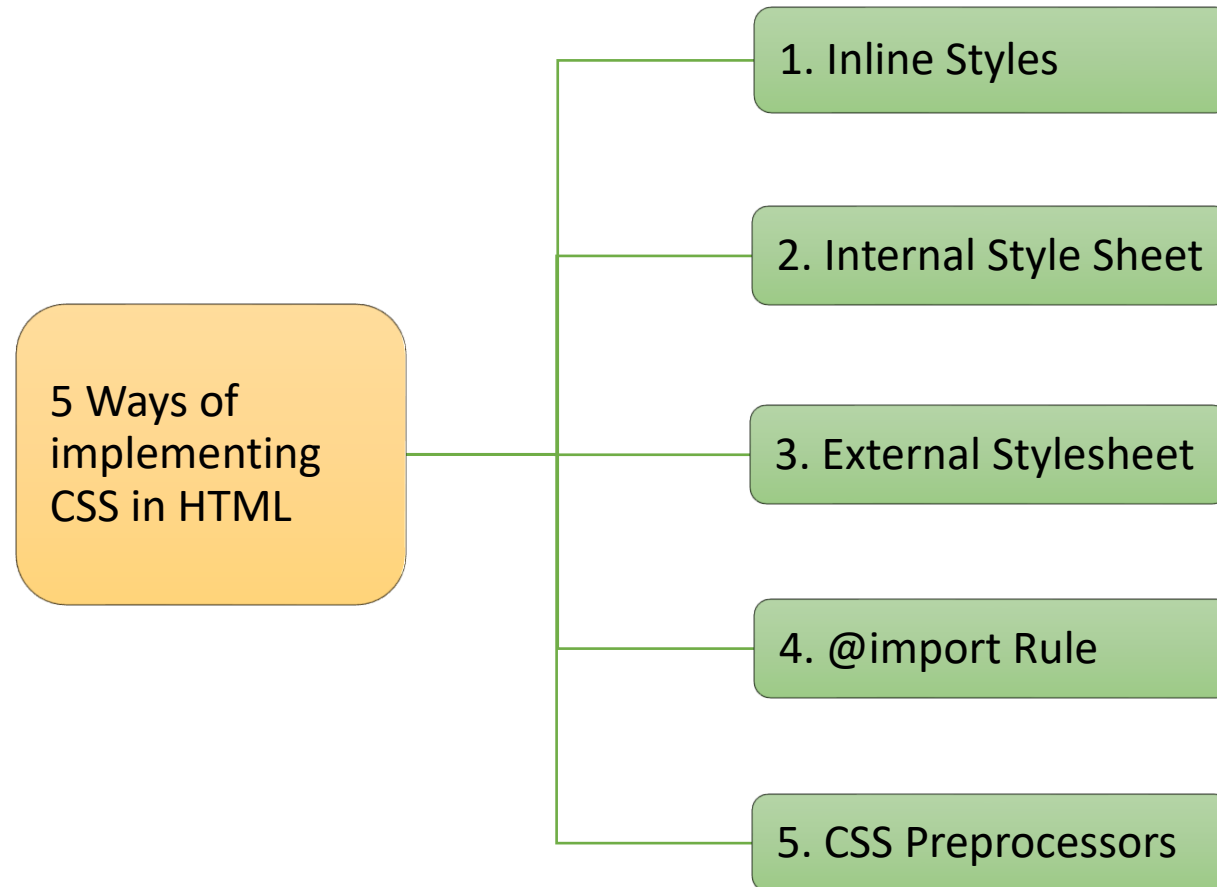
```
<html>
  <head>
    <meta charset="UTF-8" />
    <title>Inline CSS Example</title>
  </head>
  <body>
    <div style="background-color: ■ #f0f0f0;">
      <h1 style="color: ■ blue">Interview</h1>
      <p style="font-size: 30px">Happy</p>
    </div>
  </body>
</html>
```

Interview

Happy

CSS

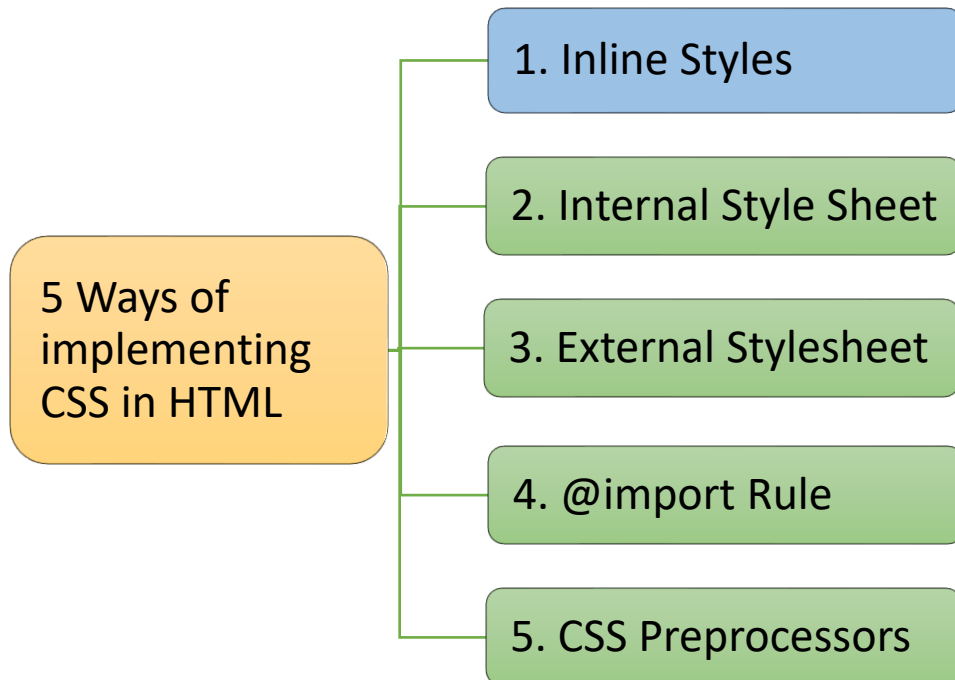
Q. What is **CSS**? What are the 5 ways to **Implement CSS** in HTML? **V. IMP.**



Q. What is **Inline Style** in CSS? When to **use** it in real applications?



- ❖ Inline Styles apply styles directly to individual HTML elements using the **style attribute**.
- ❖ This method is suitable for applying styles to a **single element**.

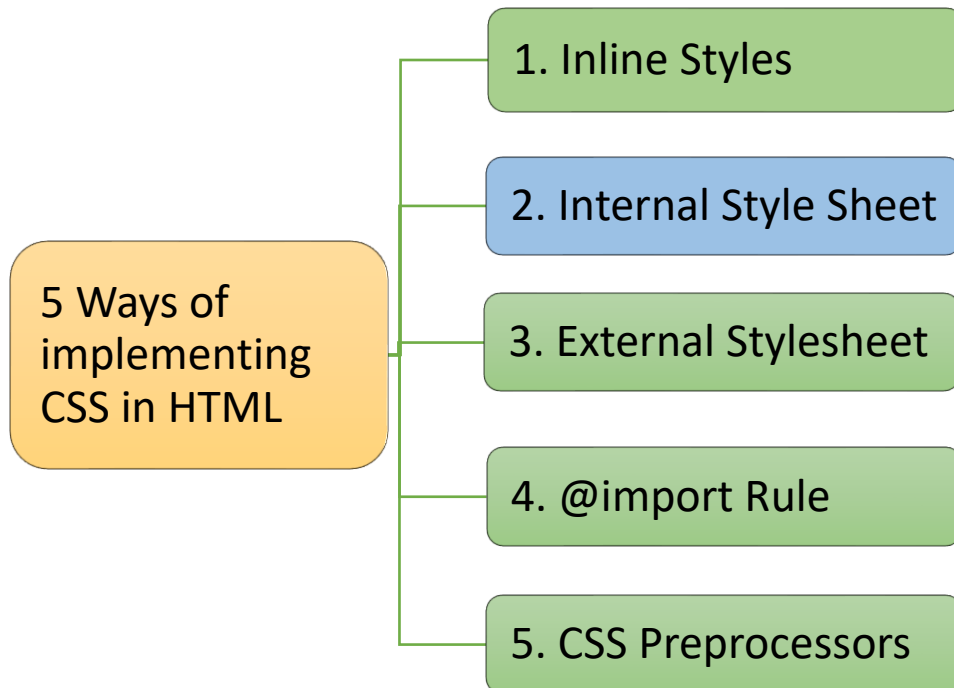


```
<html>
  <head> </head>
  <body>
    <h1 style="text-align: center">Inline Styles</h1>
    <p style="color: green">Paragraph</p>
  </body>
</html>
```

Q. What is **Internal Stylesheet** in CSS? When to **use** it in real applications?



- ❖ Internal Style Sheets can be implemented by adding the **<style> element** in the <head> section of HTML.
- ❖ This method is suitable for **smaller projects**.



```
<html>
  <head>
    <style>
      div {
        text-align: left;
      }
      .para {
        color: green;
      }
    </style>
  </head>
  <body>
    <div>Internal Style Sheet</div>
    <p class="para">Paragraph</p>
  </body>
</html>
```

Q. What is **External Stylesheet** in CSS? When to **use** it in real applications? **V. IMP.**



- ❖ In External Stylesheet, a **separate CSS file** is created(.css extension) and link it in the HTML file using the **<link> element**.
- ❖ This is most common and **recommended**.

5 Ways of
implementing
CSS in HTML

1. Inline Styles

2. Internal Style Sheet

3. External Stylesheet

4. @import Rule

5. CSS Preprocessors

```
# 2-CssTypes3.css > ...  
1  h1 {  
2    | text-align: center;  
3  }  
4  p {  
5    | color: green;  
6  }
```

```
<html>  
  <head>  
    | <link rel="stylesheet" href="2-CssTypes3.css">  
  </head>  
  <body>  
    | <h1>Internal Style Sheet</h1>  
    | <p>Paragraph</p>  
  </body>  
</html>
```

Q. What are the 5 advantages of External Stylesheet? V. IMP.



5 Advantages of CSS:

1. **Separation** of content(HTML) and presentation(Style).

2. **Reusability** of CSS classes in multiple elements.

3. Keeps things organized and **structured**.

4. Adapts to Different **Devices**.

5. Improves Website **Speed**.

```
1-Css2.css > ...
1  .container {
2  |   background-color: #f0f0f0;
3  |   padding: 20px;
4  | }
5  .heading {
6  |   color: #333;
7  | }
8  .paragraph {
9  |   font-family: Arial, sans-serif;
10 | }
```

Q. How do you **Include CSS** in a webpage or HTML?



- ❖ 1. External Stylesheet: By using the **<link> element** in the **<head>** section of the HTML document.

```
<head>
  <title>Sass Example</title>
  <link rel="stylesheet" href="4-PreProcessor.css" />
</head>
```

- ❖ 2. Internal Style Sheet: By using the **<style> element** in the **<head>** section.

```
<head>
  <style>
    h1 {
      text-align: center;
    }
  </style>
</head>
```

- ❖ 3. Inline Styles: Apply styles directly to individual HTML elements using the **style attribute**.

```
<body>
  <h1 style="text-align: center">Inline Styles</h1>
  <p style="color: green">Paragraph</p>
</body>
```

Q. How to implement CSS using @import rule?



- ❖ @import rule is to include **external CSS** file to main CSS file.
- ❖ This method is less commonly used due to potential **performance issues**.

```
/* 3-Main.css */
@import url("3-External.css");

p {
  font-size: 18px;
}
```

```
/* 3-External.css */
h1 {
  color: blue;
}
```

5 Ways of
implementing
CSS in HTML

1. Inline Styles

2. Internal Style Sheet

3. External Stylesheet

4. @import Rule

5. CSS Preprocessors

```
<html>
  <head>
    <link rel="stylesheet" href="3-Main.css" />
  </head>
  <body>
    <h1>Imported Styles</h1>
    <p>Paragraph</p>
  </body>
</html>
```

Q. What is **CSS Preprocessors**? What is **SCSS**? **V. IMP.**

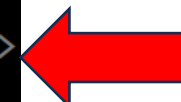


- ❖ CSS preprocessors are **scripting languages** that provide additional syntax and features that are not available in traditional CSS. For example, variables, loops, and conditional statements.
- ❖ SCSS is a type of CSS preprocessor.

```
4-PreProcessor.scss > ...  
1  
2 $primary-color: blue;  
3 $base-font-size: 16px;  
4  
5 .heading {  
6     font-size: $base-font-size * 1.2;  
7     color: $primary-color;  
8 }
```

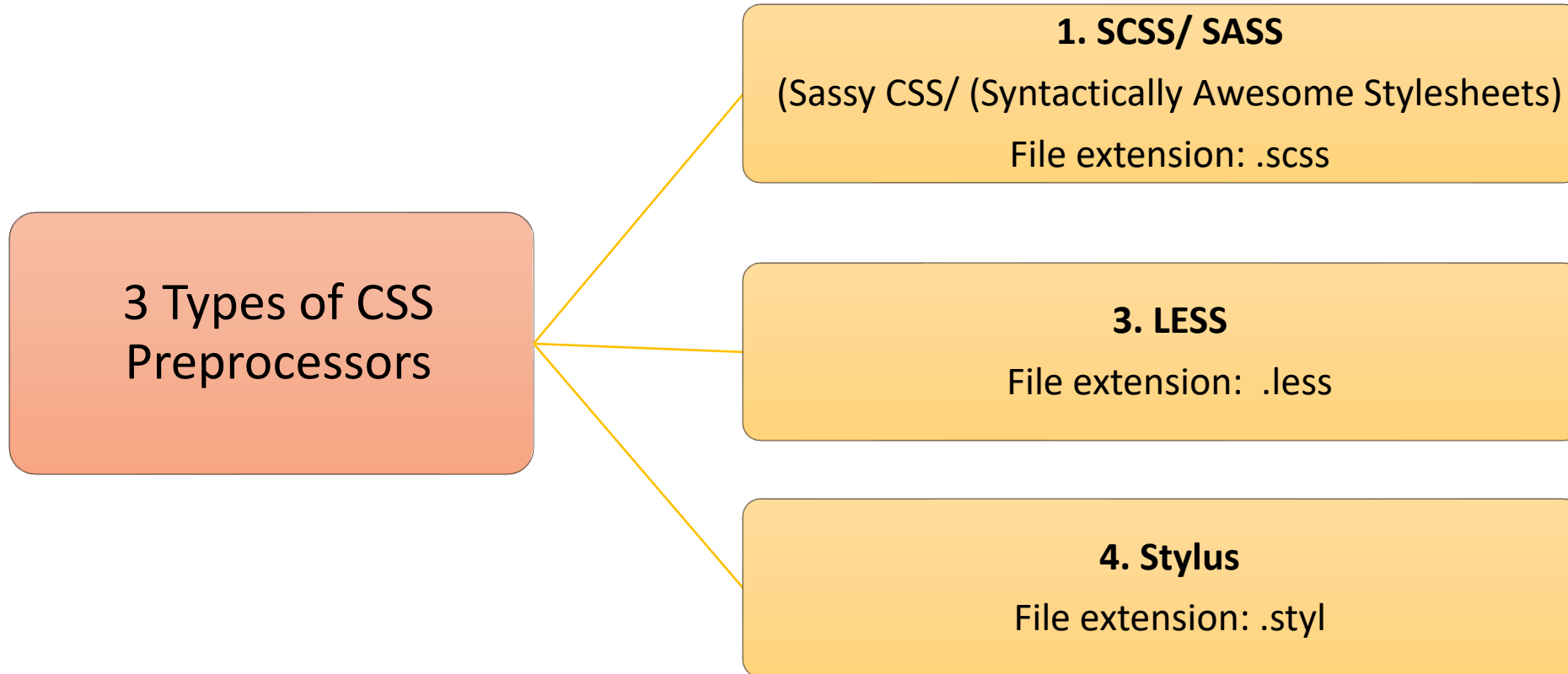


```
# 4-PreProcessor.css > ...  
1  
2 /* Generated css from scss */  
3  
4 .heading {  
5     font-size: 19.2px;  
6     color: blue;  
7 }
```



```
<html>  
  <head>  
    <title>Sass Example</title>  
    <link rel="stylesheet" href="4-PreProcessor.css" />  
  </head>  
  <body>  
    <h1 class="heading">Interview Happy</h1>  
    <p class="paragraph">Crack Interviews</p>  
  </body>  
</html>
```

Q. What are the 3 Types of CSS Preprocessors?



2. Selectors in CSS

Q. What are **Selectors** in CSS? How many types of selectors are there?

Q. Difference between **ID**, **Element** & **Class** selector? When to use which selector?

Q. What are **Universal selectors**?

Q. What are the **Descendant Selectors** in CSS?

Q. What are **Attribute Selectors** in CSS? What are its 5 Types?

Q. What are **Child Selectors** in CSS?

Q. How Child Selector is different from Descendent selectors?

Q. What are **Pseudo-class Selector & Pseudo-element Selector**?

Q. What are **Selectors** in CSS? How many **Types** of selectors are there? **V. IMP.**



- ❖ In CSS, selectors are patterns that are used to select and style HTML elements.

Types of Selectors

1. ID Selector(#)
2. Element Selector(no symbol)
3. Class Selector(.)
4. Universal Selector(*)
5. Descendant selector
6. Child Selector(>)

```
<html>
  <head>
    <link rel="stylesheet" href="3-Main.css" />
  </head>
  <body>
    <h1>Imported Styles</h1>
    <p>Paragraph</p>
  </body>
</html>
```



```
h1 {
  color: blue;
}
```

Q. Difference between ID, Element & Class selector? When to use which selector? V. IMP.



```
/* ID Selector */  
<div id="my-id">  
| <p>This background will be yellow.</p>  
</div>
```

```
/* # ID Selector */  
#my-id {  
| background-color: ■yellow;  
}
```

This background will be yellow.

```
<h2>This text font size will be 20px.</h2>
```

```
/* Element Selector */  
h2 {  
| font-size: 20px;  
}
```

This text font size will be 20px.

```
<div class="my-class">  
| <p>This text will be blue.</p>  
</div>
```

```
/* . Class Selector */  
.my-class {  
| color: ■blue;  
}
```

This text will be blue.

Q. Difference between ID, Element & Class selector? When to use which selector? V. IMP.



1. ID Selector (#)

- ID selectors are used for unique elements on a page, and they should only be assigned to one element.

```
/* # ID Selector */
#my-id {
| background-color: ■yellow;
}
```

This background will be yellow.

2. Element Selector

- Element selectors are used when you want to apply a style to all elements of a specific type.

```
/* Element Selector */
h2 {
| font-size: 20px;
}
```

This text font size will be 20px.

3. Class Selector (.)

- Class selectors are used when you want to apply a style to multiple elements with the same class.

```
/* . Class Selector */
.my-class {
| color: ■blue;
}
```

This text will be blue.

Q. What are **Universal** selectors?



- ❖ Universal selector applied the specified styles to all elements, regardless of their type or attributes.

```
<body>
  <div id="my-id">
    | <p>Interview</p>
  </div>

  <h2>Happy</h2>
</body>
```

Interview

Happy

```
/* * Universal Selector */
* {
  | color: ■ blue;
  | font-size: 24px;
  }
}
```

Q. What are the **Descendant Selectors** in CSS?



- ❖ A descendant selector in CSS is a selector that targets an element that is a descendant of another specified element(**nested elements**).

```
<body>
  <div class="container">
    <p>Paragraph 1</p>
    <p>Paragraph 2</p>
    <h4>No css applied</h4>
  </div>

  <ul>
    <li>Item 1</li>
    <li>Item 2</li>
  </ul>
</body>
```

```
.container p {
  color: blue;
  font-size: 16px;
}

ul li {
  background-color: red;
  padding: 5px;
  margin: 5px;
}
```

Paragraph 1

Paragraph 2

- Item 1

- Item 2

Q. What are **Attribute Selectors** in CSS? What are its **5 Types**?



- ❖ Attribute selectors in CSS target HTML elements based on their attributes and their values.

```
<input type="text"/>
<input type="password"/>
<a href="https://www.example.com">Link 1</a>
<a href="http://www.example.pdf">Link 2</a>
```

		Link 1 Link 2
----------------------	--------------------------	---

```
/* Attribute Exists Selector */
input[type] {
  border: 1px solid black;
}

/* Attribute Equals Selector */
a[href="https://www.example.com"]
{
  color: blue;
}

/* Attribute Contains Selector */
input[type*="text"] {
  background-color: #f0f0f0;
}

/* Attribute Starts With Selector */
a[href^="https://"]
{
  color: green;
}

/* Attribute Ends With Selector */
a[href$=".pdf"] {
  text-decoration: underline;
}
```

Q. What are **Child Selectors** in CSS?



- ❖ A child selector targets an element that is a **immediate child** of specified parent element.
- ❖ It's denoted by the **>** symbol.

```
<body>
  <div class="parent">
    <span>Direct Child Span</span>
    <div>
      <span>Not Direct Child Span</span>
    </div>
  </div>
</body>
```

```
/* Style applied to direct children
of class "parent" that are spans */
.parent > span {
  color: green;
}
```

Direct Child Span
Not Direct Child Span

Q. How Child Selector is **different** from Descendant selectors?



- ❖ The child selector targets **immediate children**, while the descendant selector targets all descendants, regardless of depth.

Q. What are **Pseudo-class Selector** and **Pseudo-element Selector**?



- ❖ A pseudo-class selector is used to define the **special state** of an HTML element, like when a user hovers over an element.
- ❖ A pseudo-element selector targets a **specific part** of an element, such as `::before` to insert content before an element.

```
<body>
  <div class="box">
    <p>Hover over me</p>
    <span>Content added before me</span>
  </div>
</body>
```

```
/* Pseudo-class Selector */
p:hover {
  color: blue;
}
```

Hover over me

```
/* Pseudo-element Selector */
span::before {
  content: "Yes: ";
  font-weight: bold;
}
```

Yes: Content added before me

3. Layout and Positioning

Q. Explain the concept of the **Box Model** in CSS.

Q. What is the difference between **Padding** & **Margin**?

Q. What is the difference between **display: inline** & **display: block**?

Q. What is the difference between **display: inline** & **display: inlineblock**?

Q. What is **z-index** property in CSS? How same z-index elements will be displayed?

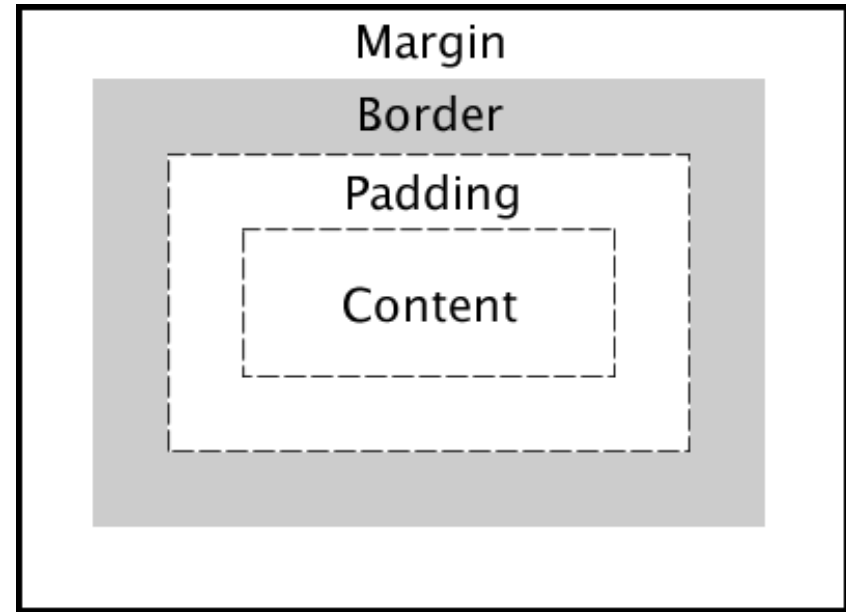
Q. What is the role of **float** property in CSS?

Q. What is the difference between **position: relative/ absolute/ fixed**?

Q. Explain the concept of the **Box Model** in CSS. **V. IMP.**



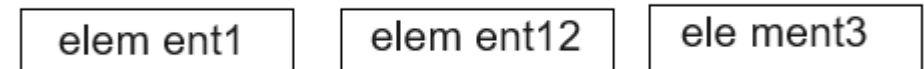
- ❖ The "box model" describes the **layout of elements** on a webpage in terms of their dimensions and spacing.
- 1. **Content:** This is the actual content of the element, such as text, images, videos, etc.
- 2. **Padding:** This is the space between the content and the border.
- 3. **Border:** This is the line that separate the content from the surrounding elements.
- 4. **Margin:** This is the space between the borders of adjacent elements. Margins help control the space between elements.
- ❖ In CSS, you can control each of these components using various properties like width, height, padding border, margin.



❖ Without box model

elem ent1 elem ent2 ele ment3

❖ With box model

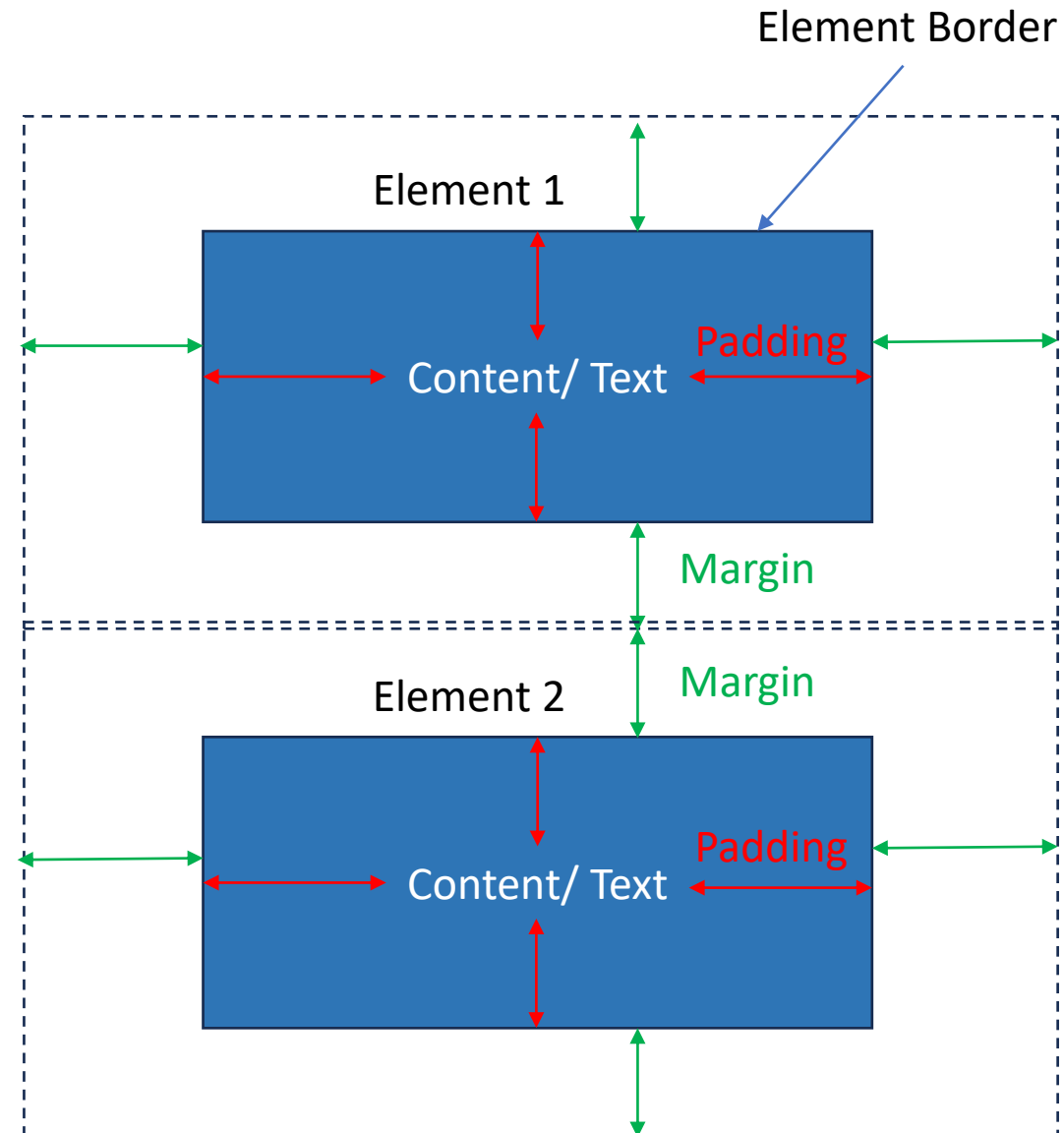


Q. What is the difference between **Padding** and **Margin**? **V. IMP.**



- ❖ Padding is the space between the content of an element and its border.
- ❖ Margin is the space outside an element, between the element and its surrounding elements.

```
.h1 {  
  border: 1px solid black;  
  margin-top: 20px;  
  margin-right: 30px;  
  margin-bottom: 40px;  
  margin-left: 50px;  
  padding-top: 10px;  
  padding-right: 20px;  
  padding-bottom: 30px;  
  padding-left: 40px;  
}
```



Q. What is the difference between **display: inline** & **display: block**? V. IMP.



- ❖ Block elements take up the **full width** of their parent container by default
- ❖ Inline elements take up as much width as per the **width of content only**.

```
<body>
  <h3>Display Properties</h3>
  <div class="block">Block Element</div>
  Text1
  <div class="inline">Inline Element</div>
  Text2
</body>
```

```
.block {
  display: block;
  border: 1px solid black;
  padding: 20px;
  margin: 20px;
}
```

Block Element

```
.inline {
  display: inline;
  border: 1px solid black;
  padding: 20px;
  margin: 20px;
}
```

Text1

Inline Element

Text2

Q. What is the difference between `display: inline` and `display: inlineblock`? V. IMP.



- ❖ Inline elements do not accept width and height values whereas inline-block accept width and height also.

```
<body>
  <div class="inline">Inline Element</div>Text2<br />
  <div class="inline-block">Inline-Block</div>Text3
</body>
```

```
.inline {
  display: inline;
  border: 1px solid black;
  height: 50px;
  width: 50px;
  margin: 50px;
}
```

Inline Element

Text2

```
.inline-block {
  display: inline-block;
  border: 1px solid black;
  height: 50px;
  width: 50px;
  margin: 50px;
}
```

Inline-Block

Text3

Q. What is **z-index** property in CSS? How **same** z-index elements will be displayed?



- ❖ z-index property in CSS is used to control the **stacking order** of positioned elements.
- ❖ Elements with a higher z-index value will be displayed on top of elements with a lower z-index value.
- ❖ If two elements have the same z-index, the one that comes later in the HTML source will appear on top.

```
<body>
  <div class="container">
    <div class="box1"></div>
    <div class="box2"></div>
  </div>
</body>
```



```
.box1 {
  position: absolute;
  width: 100px;
  height: 100px;
  background-color: red;
  z-index: 1;
}

.box2 {
  position: absolute;
  width: 100px;
  height: 100px;
  background-color: blue;
  z-index: 2;
}
```

Q. What is the role of **float property** in CSS?



- ❖ The float property in CSS is used to specify how an element should align or "float" within its parent container.

```
<body>
  <div class="sidebar">
    <h2>Sidebar</h2>
  </div>

  <div class="main-content">
    <h1>Main Content</h1>
  </div>
</body>
```

```
.sidebar {
  width: 30%;
  float: right;
  background-color: lightblue;
  padding: 20px;
  box-sizing: border-box;
}

.main-content {
  width: 70%;
  padding: 20px;
  box-sizing: border-box;
  background-color: lightgreen;
}
```

Main Content

Sidebar

Q. What is the difference between **position: relative/ absolute/ fixed**? **V. IMP.**



- ❖ position: relative positions an element relative to its normal position of the page.
- ❖ position: absolute is positioned as per the parent element.
- ❖ position: fixed will fix the position of element even if the page is scrolled.

```
<body>
  <div class="relative-box">
    <h2>Relative Box</h2>
  </div>

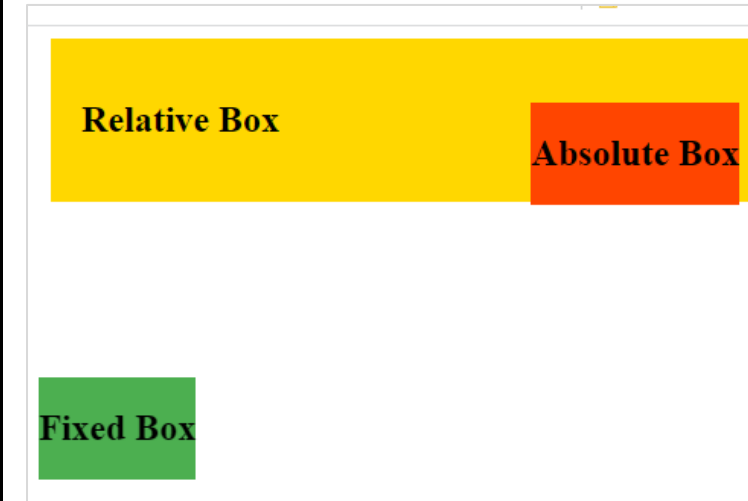
  <div class="absolute-box">
    <h2>Absolute Box</h2>
  </div>

  <div class="fixed-box">
    <h2>Fixed Box</h2>
  </div>
</body>
```

```
.relative-box {
  position: relative;
  left: 20px;
  padding: 20px;
  background-color: #ffd700;
}

.absolute-box {
  position: absolute;
  top: 50px;
  right: 20px;
  background-color: #ff4500;
}

.fixed-box {
  position: fixed;
  bottom: 50px;
  left: 20px;
  background-color: #4caf50;
}
```



4. Responsive Design, Flexbox & Grid Layout

Q. What is the role of **meta viewport** tag?

Q. What are **Media Queries** in CSS?

Q. What is **display: flex** in CSS?

Q. What is CSS **Grid Layout** in CSS?

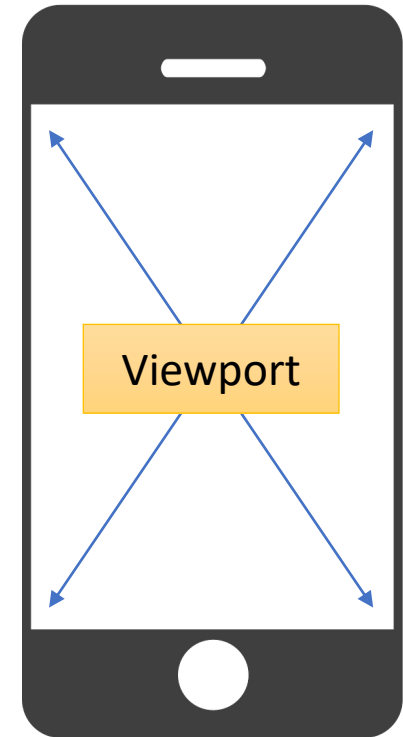
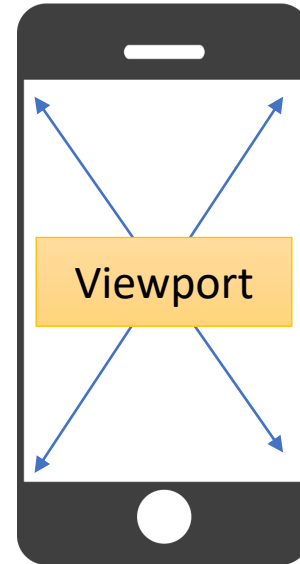
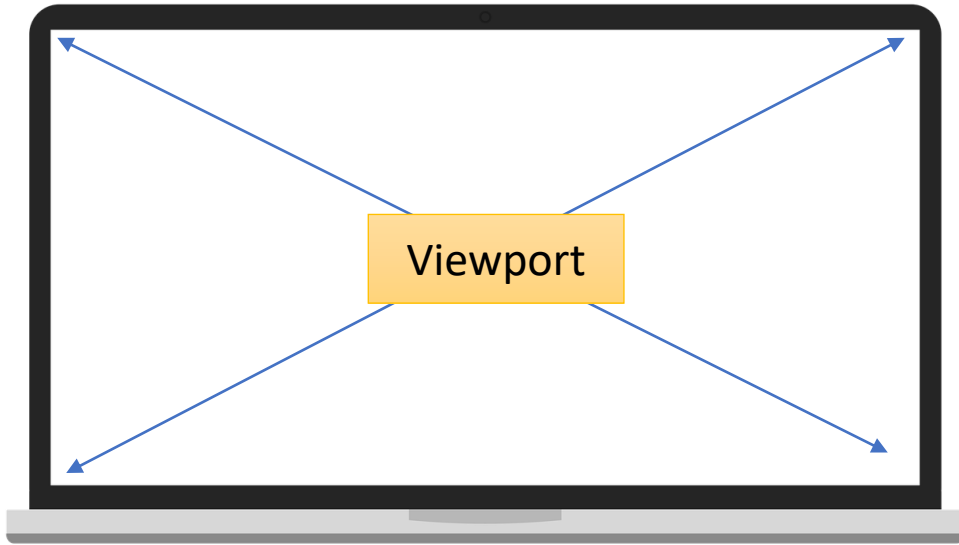
Q. How can you create a **Sticky Header** in CSS?

Q. What are the 5 best practices for creating a Responsive Design using CSS?

Q. What is the role of **meta viewport** tag? **V. IMP.**



- ❖ In web design, the "viewport" is the portion of the webpage that the user can see on their screen at any given time, without scrolling.



Q. What is the role of **meta viewport** tag? **V. IMP.**



- ❖ The <meta name="viewport"> tag in HTML is used to adjust the webpage on different devices and screen sizes.

❖ Name attribute specifies that the meta tag is a meta viewport.

❖ Content width property tells the browser to set the width of the viewport to the width of the device's screen.

❖ Content initial-scale property sets the initial zoom level when the page is first loaded. A value of 1.0 means that the page is displayed at its actual size without any zooming.

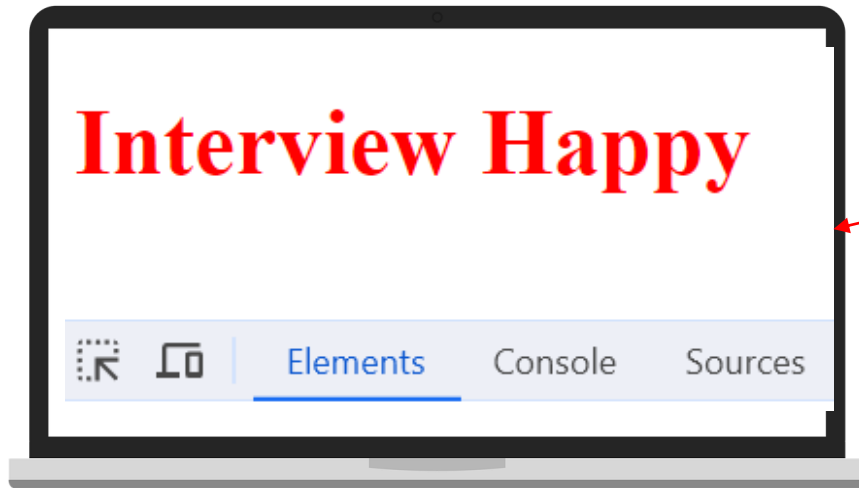
```
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <link rel="stylesheet" href="18-MediaQueries.css" />
</head>
```

Q. What are **Media Queries** in CSS? **V. IMP.**

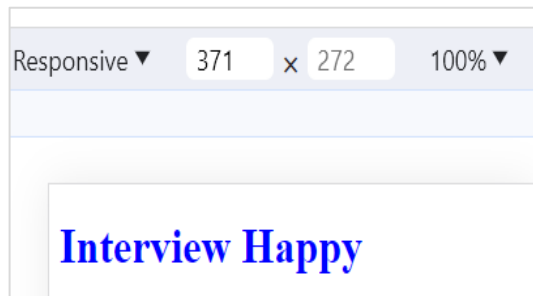


- ❖ Media queries in CSS allows you to apply different styles to elements based on the size of device screen.

```
<body>
  <h1>Interview Happy</h1>
</body>
```



```
h1 {
  color: red;
}
@media screen and (max-width: 768px) {
  h1 {
    color: blue;
  }
}
```



Q. What is **display: flex** in CSS?



- ❖ `display: flex` turns an element into a container that structure and align its child elements in a better way. It enables a flexbox layout for the container.
- ❖ `justify-content` property aligns flex items along the main axis of the flex container. It controls the spacing between and around items.

[Home](#) [About](#) [Services](#) [Contact](#)



[Home](#)

[About](#)

[Services](#)

[Contact](#)



```
<body>
  <nav class="nav-container">
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Services</a>
    <a href="#">Contact</a>
  </nav>
</body>
```

```
.nav-container {
  display: flex;
  justify-content: space-around;
  background-color: #333;
  padding: 10px;
}

.nav-container a {
  color: white;
}
```

Q. What is **CSS Grid Layout** in CSS?



- ❖ The CSS Grid Layout Module offers a grid-based layout system, with rows and columns.

```
<body>
  <div class="grid-container">
    <div>1</div>
    <div>2</div>
    <div>3</div>
    <div>4</div>
    <div>5</div>
    <div>6</div>
  </div>
</body>
```

Without grid css

1
2
3
4
5
6

With grid css

```
.grid-container {
  display: grid;
  grid-template-columns: auto auto auto;
}
```

1	2	3
4	5	6

Q. How can you create a **Sticky Header** in CSS?



- ❖ Position: fixed can be used to implement sticky header in CSS.

```
<body>
  <div class="header">
    <h2>My Header</h2>
  </div>
  <div class="content"></div>
</body>
```

```
.header {
  position: fixed;
  padding: 10px 16px;
  background: lightblue;
}
.content {
  height: 2000px;
}
```

My Header

Q. What are the 5 best practices for creating a **Responsive Design** using CSS? **V. IMP.**



5 Best practices for creating responsive design

1. Use Responsive Units(%)

2. Use Media Queries

3. Flexible Layout

4. Images and Media size

5. Viewport Meta Tag

```
.container {  
  width: 80%;  
}  
  
@media screen and (min-width: 768px) {  
  .header {  
    font-size: 24px;  
  }  
}  
  
.container {  
  display: flex;  
}  
  
img {  
  max-width: 100%;  
  height: auto;  
}
```

Q. What are the 5 best practices for creating a **Responsive Design** using CSS? **V. IMP.**



1. Use Responsive Units:

Use relative units like percentages (%) or em units for sizes (width, height, padding, margin, etc.) instead of fixed units like pixels (px).

2. Use Media Queries:

Media queries allow you to apply different styles based on the characteristics of the device, such as screen width, height, and orientation. They are essential for creating a responsive design.

3. Flexible Grid Systems:

Use CSS Grid or Flexbox to create flexible layouts that can adapt to different screen sizes and orientations.

4. Images and Media:

Use max-width: 100%; on images to ensure they don't overflow their containers. This allows images to scale down proportionally on smaller screens.

5. Viewport Meta Tag:

Include the viewport meta tag in the <head> of your HTML to control how the page is displayed on mobile devices. It sets the initial scale, width, and zoom behavior.

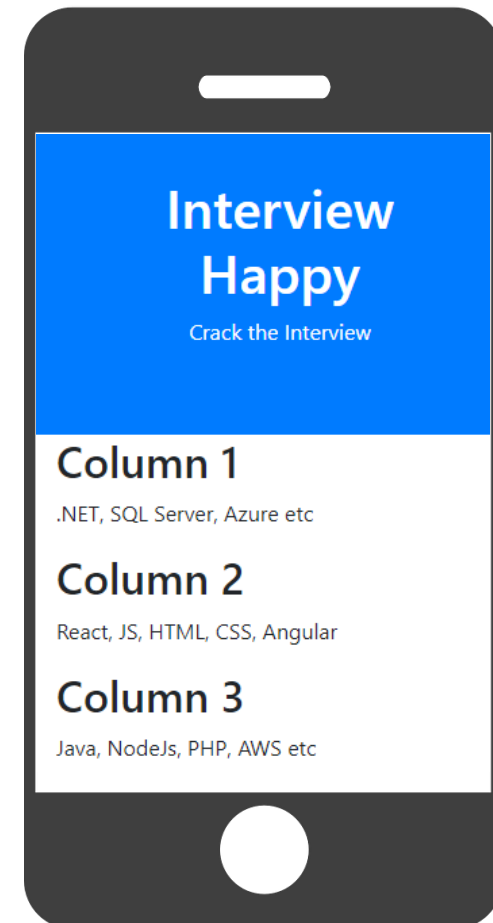
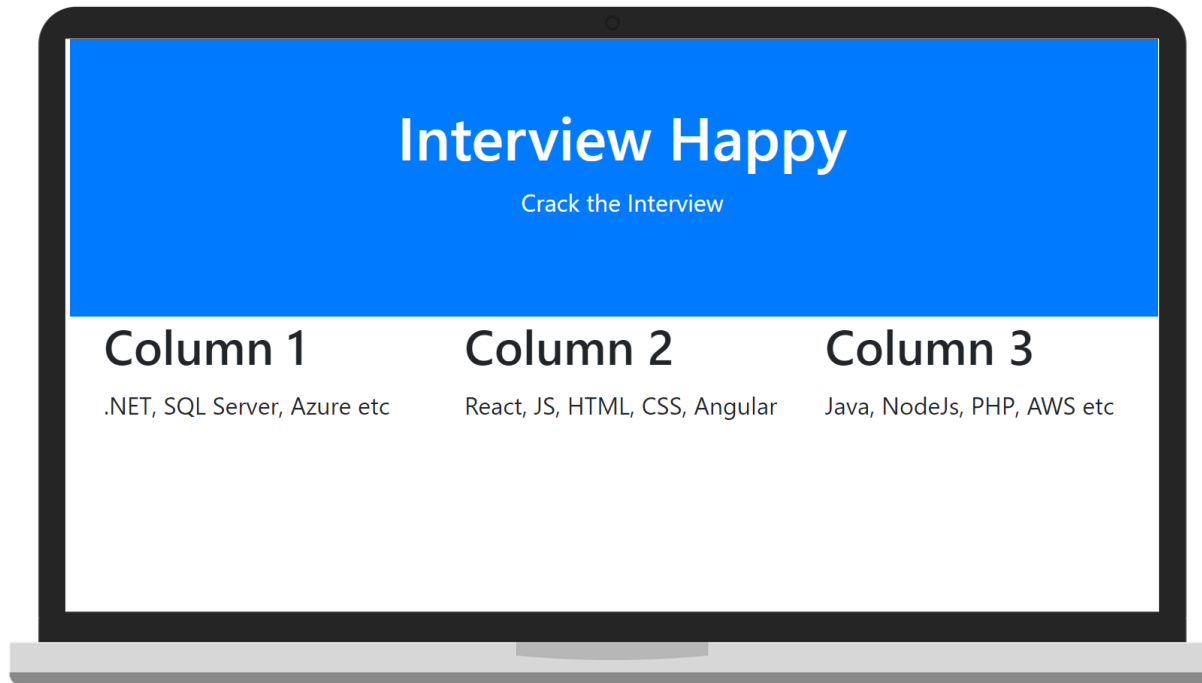
1. Bootstrap

- Q. What is **Bootstrap**? What are the other 5 responsive design frameworks?
- Q. What are the **5 Advantages** of using Bootstrap?
- Q. What are the 2 ways to include Bootstrap framework for your website.
- Q. Explain the **Grid System** in Bootstrap?
- Q. What is the difference between **col-xs, col-sm, col-md, col-lg & col-xl**?
- Q. What are Bootstrap **Components**? What are Top 10 bootstrap components?
- Q. What is a Bootstrap **Modal** component?
- Q. What is a Bootstrap **Navigation** component?
- Q. What is Bootstrap **Carousel** component?
- Q. Explain the difference between Bootstrap's **container & container-fluid**?

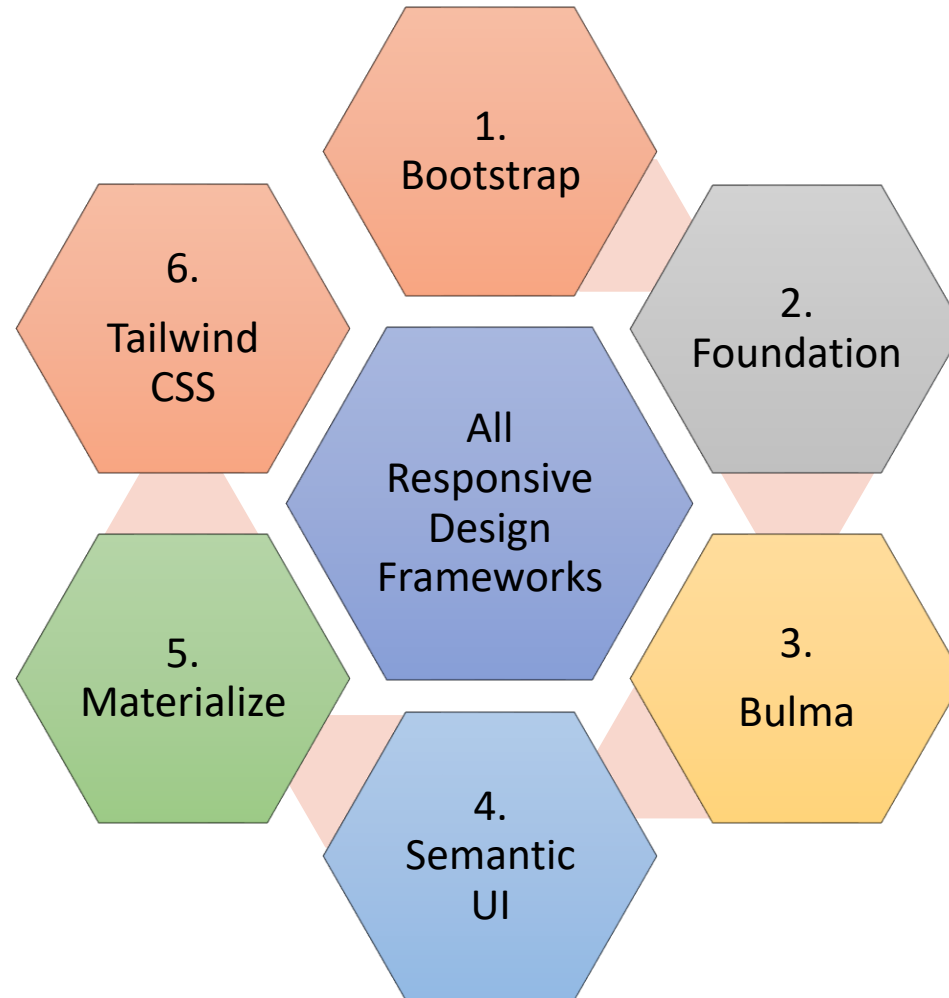
Q. What is **Bootstrap**? What are the other **5** responsive design frameworks? **V. IMP.**



- ❖ Bootstrap is a popular open-source front-end framework which provides responsive and mobile-first CSS.



Q. What is **Bootstrap**? What are the other **5** responsive design frameworks? **V. IMP.**



Q. What are the 5 Advantages of using Bootstrap?



1. Rapid Development

- Bootstrap provides a wide range of ready-to-use components like navigation bars, buttons, forms, modals, and more. This saves time and effort in writing code from scratch.

2. Responsive Design

- Bootstrap is built with a mobile-first approach, meaning it prioritizes designing for smaller screens first and then scales up for larger screens. This ensures websites with Bootstrap is responsive for every device.

3. Cross-Browser Compatibility

- Bootstrap provides a consistent set of styles and components across different browsers, which helps in reducing cross-browser compatibility issues.

4. Large Community and Support

- This means there are plenty of resources, documentation, tutorials, and forums available for assistance.

5. Accessibility

- Bootstrap incorporates best practices for web accessibility, making it easier to create websites that are usable by people with disabilities.

Q. What are the 2 ways to include Bootstrap framework for your website.



2 ways to include Bootstrap:

1. Put the link of Bootstrap CDN (Content Delivery Network) in head section.

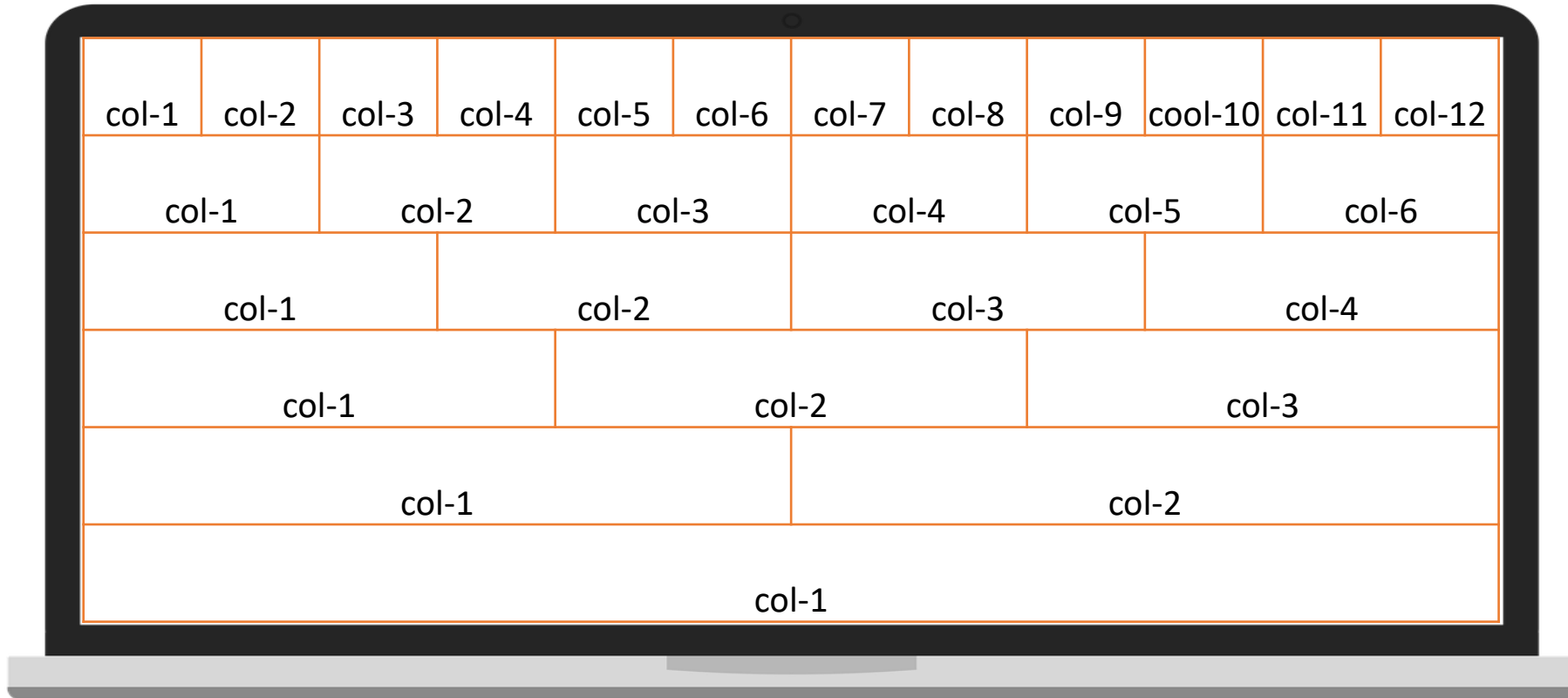
2. Download the Bootstrap files and add them locally in your HTML document.

```
<head>
  <title>Bootstrap</title>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <!-- Add Bootstrap CSS via CDN -->
  <link rel="stylesheet"
    | href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css"
  />
</head>
```

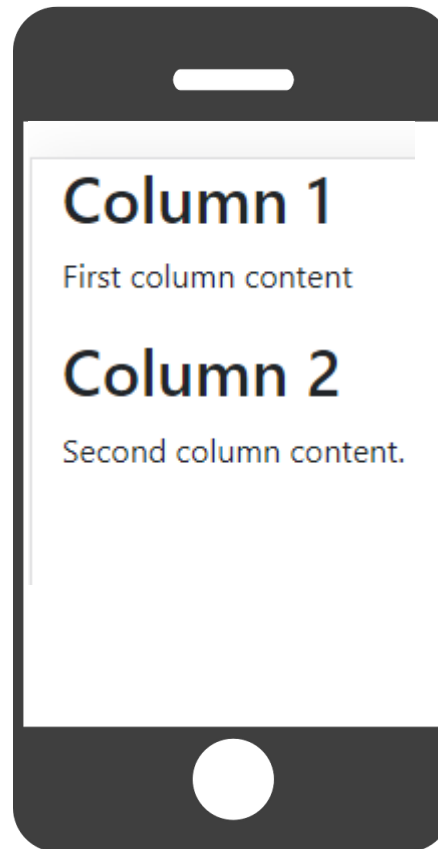
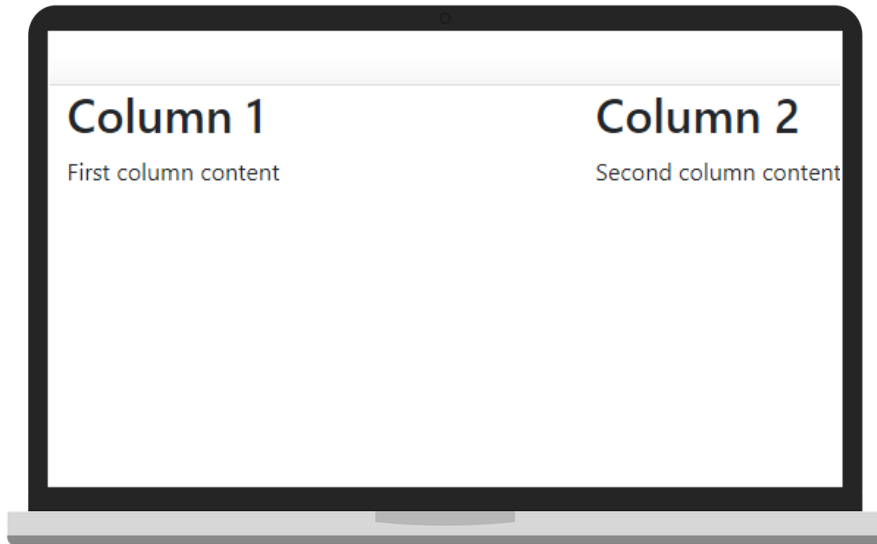
Q. Explain the **Grid System** in Bootstrap? **V. IMP.**



- ❖ Grid system is 12-column layout and is designed to adapt to various screen sizes.



Q. Explain the **Grid System** in Bootstrap? **V. IMP.**



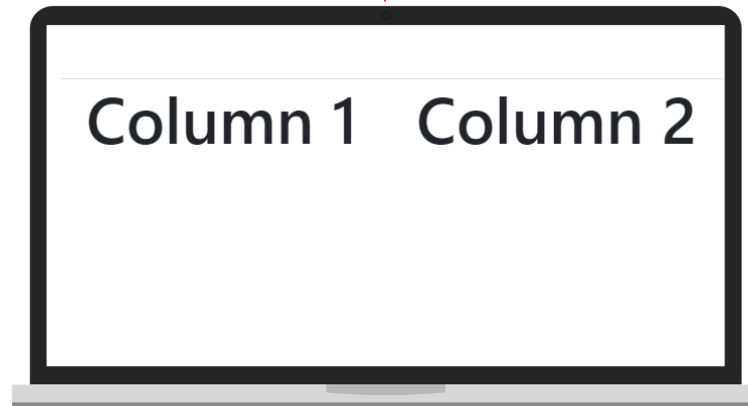
```
<body>
  <div class="container">
    <div class="row">
      <div class="col-md-6">
        <h2>Column 1</h2>
        <p>First column content</p>
      </div>
      <div class="col-md-6">
        <h2>Column 2</h2>
        <p>Second column content.</p>
      </div>
    </div>
  </div>
</body>
```

Q. What is the difference between **col-xs**, **col-sm**, **col-md**, **col-lg** & **col-xl**?

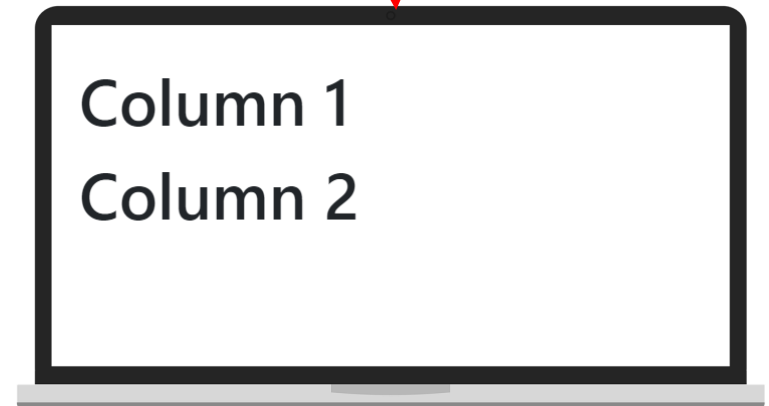


1. col-xs (Extra Small): Applies to extra small screens (phones).
2. col-sm (Small): Applies to small screens (tablets).
3. col-md (Medium): Applies to medium screens (laptops).
4. col-lg (Large): Applies to large screens (larger desktops).
5. col-xl (Extra Large): Applies to extra large screens (large TVs).

```
<div class="container">
  <div class="row">
    <div class="col-md-6">
      <h2>Column 1</h2>
    </div>
    <div class="col-md-6">
      <h2>Column 2</h2>
    </div>
  </div>
</div>
```



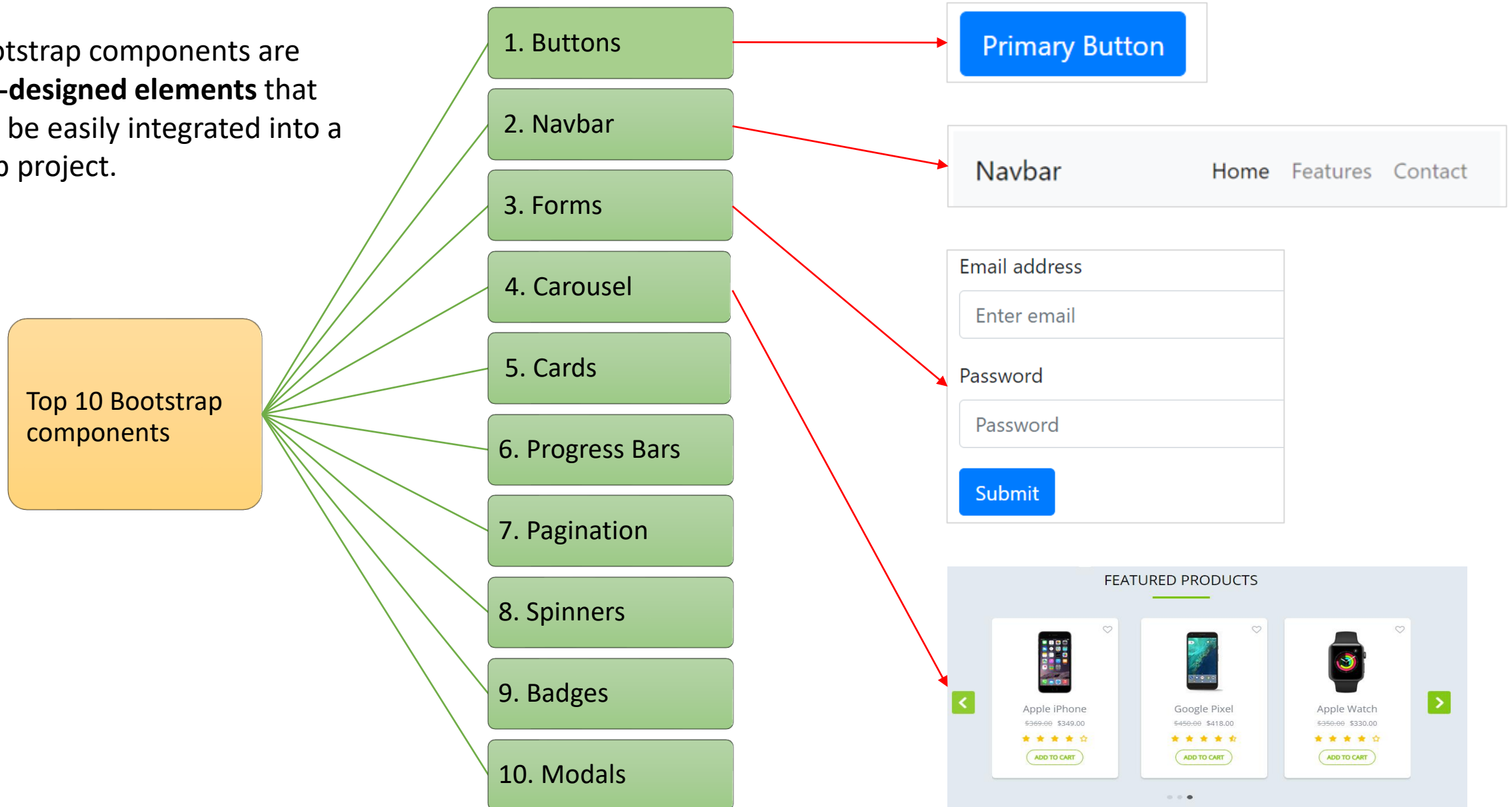
```
<div class="container">
  <div class="row">
    <div class="col-xl-6">
      <h2>Column 1</h2>
    </div>
    <div class="col-xl-6">
      <h2>Column 2</h2>
    </div>
  </div>
</div>
```



Q. What are **Bootstrap Components**? What are **Top 10** bootstrap components? **V. IMP.**



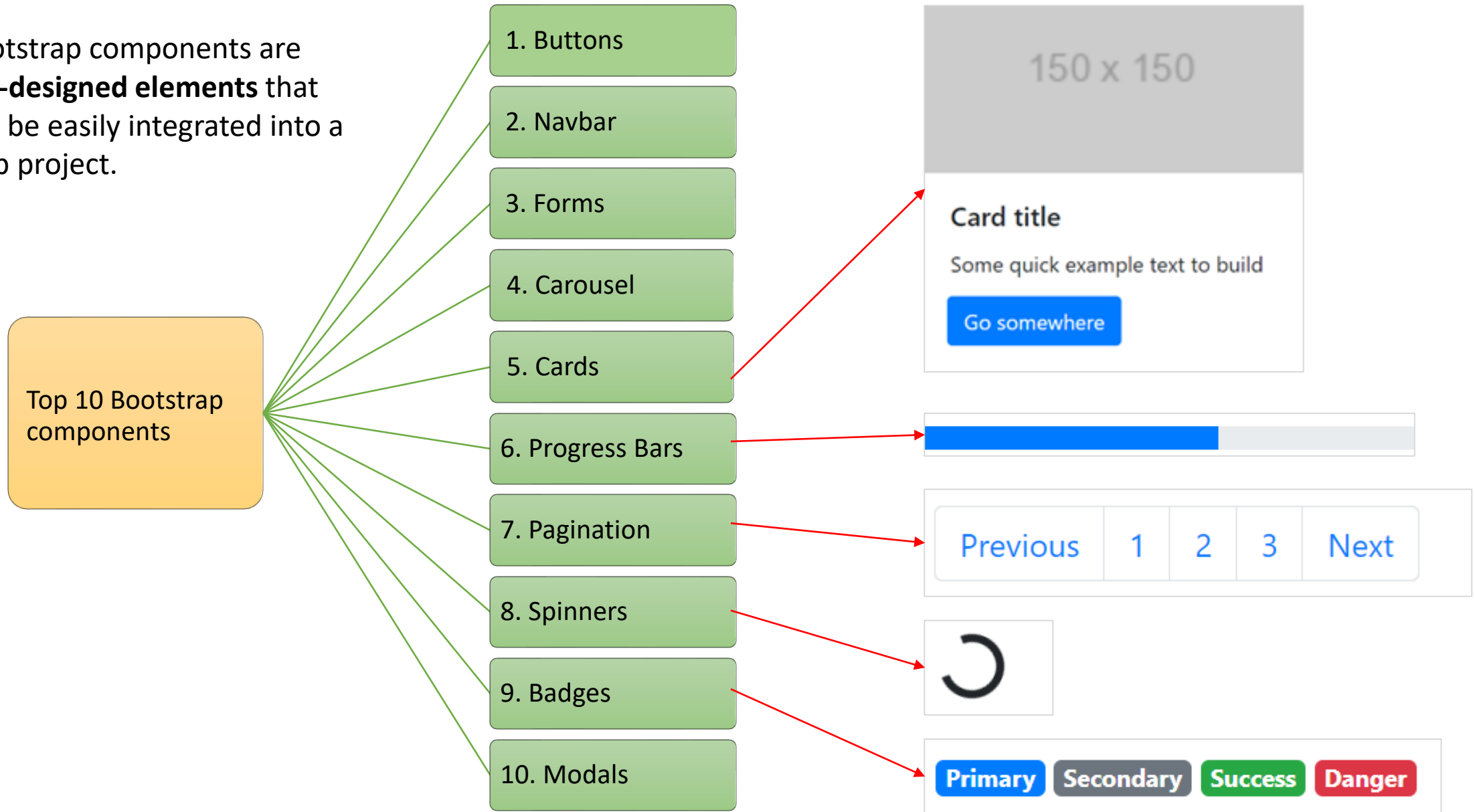
- ❖ Bootstrap components are **pre-designed elements** that can be easily integrated into a web project.



Q. What are **Bootstrap Components**? What are **Top 10** bootstrap components? **V. IMP.**



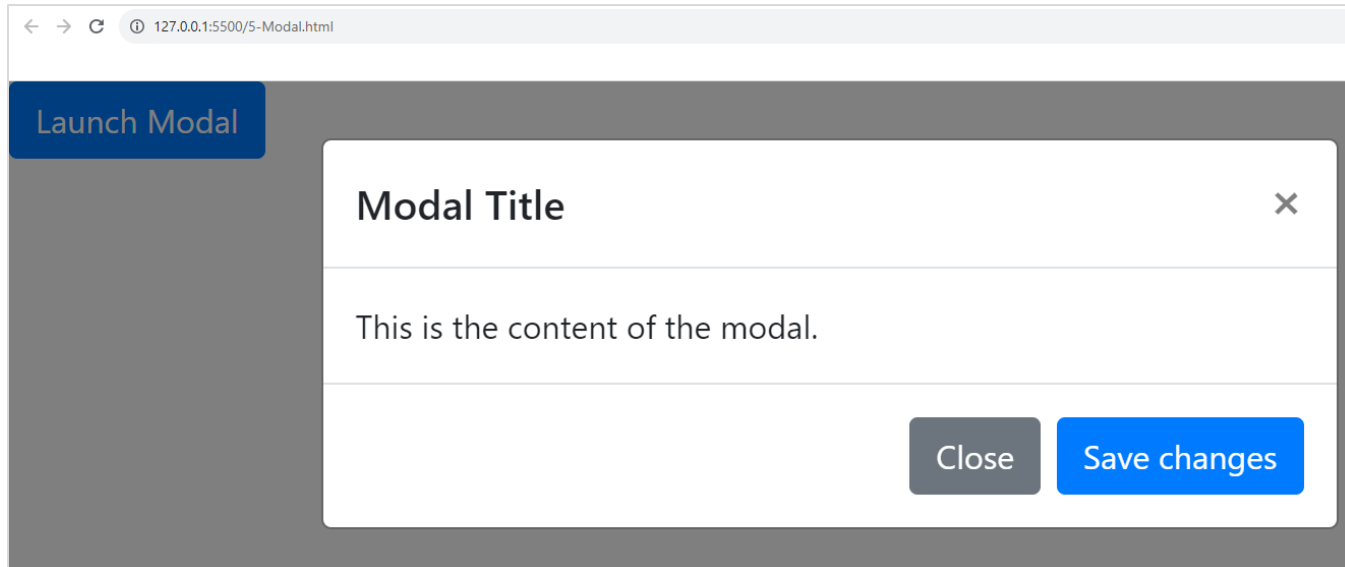
- ❖ Bootstrap components are **pre-designed elements** that can be easily integrated into a web project.



Q. What is a Bootstrap **Modal** component?



- ❖ A Bootstrap modal is a lightweight, customizable, and interactive dialog box or popup window that appears on top of the main content of a web page.



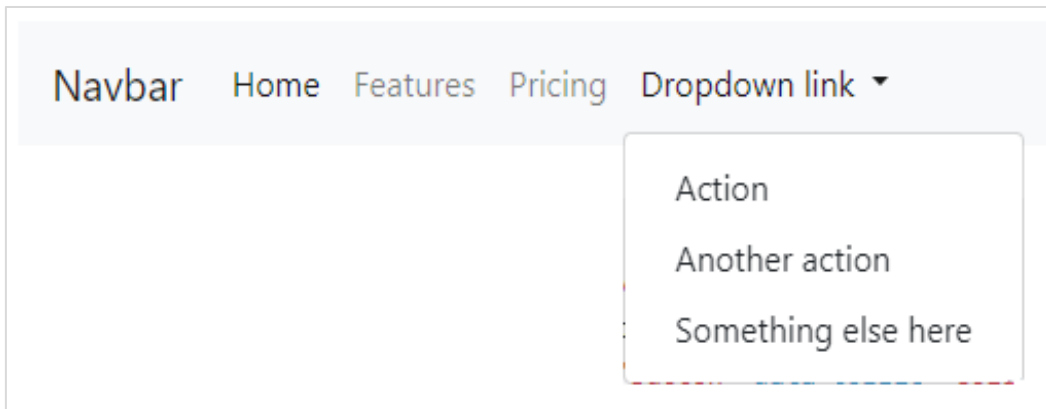
```
<body>
  <!-- Button to trigger the modal -->
  <button type="button" class="btn btn-primary" data-toggle="modal" data-target="#exampleModal">
    Launch Modal
  </button>

  <!-- Modal content -->
  <div class="modal fade" id="exampleModal" tabindex="-1">
    <div class="modal-dialog">
      <div class="modal-content">
        <div class="modal-header">
          <h5 class="modal-title" id="exampleModalTitle">Modal Title
          <button type="button" class="close" data-dismiss="modal">
            <span aria-hidden="true">&times;</span>
          </button>
        </div>
        <div class="modal-body">
          This is the content of the modal.
        </div>
        <div class="modal-footer">
          <button type="button" class="btn btn-secondary" data-dismiss="modal">Close
          <button type="button" class="btn btn-primary" data-dismiss="modal">Save changes
        </div>
      </div>
    </div>
  </div>
```

Q. What is a Bootstrap **Navigation** component?



- ❖ Bootstrap's Navigation component provides a set of styles and components to create various types of navigation menus on a webpage.

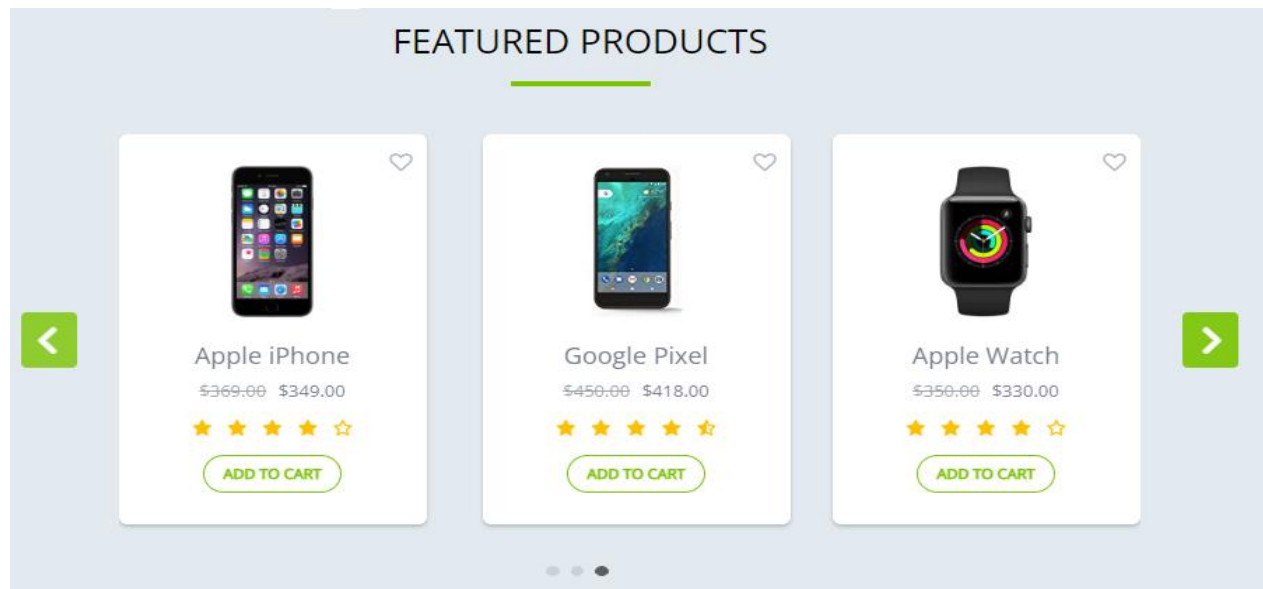


```
<body>
  <nav class="navbar" navbar-expand-lg navbar-dark">
    <a class="navbar-brand" href="#">Navbar</a>
    <button
      class="navbar-toggler"
      type="button"
      data-toggle="collapse"
      data-target="#navbarNav"
      aria-controls="navbarNav"
      aria-expanded="false"
      aria-label="Toggle navigation"
    >
      <span class="navbar-toggler-icon"></span>
    </button>
```

Q. What is Bootstrap **Carousel** component?



- ❖ Bootstrap's carousel component is a slideshow or image slider that allows you to display a series of images or content in a rotating manner.



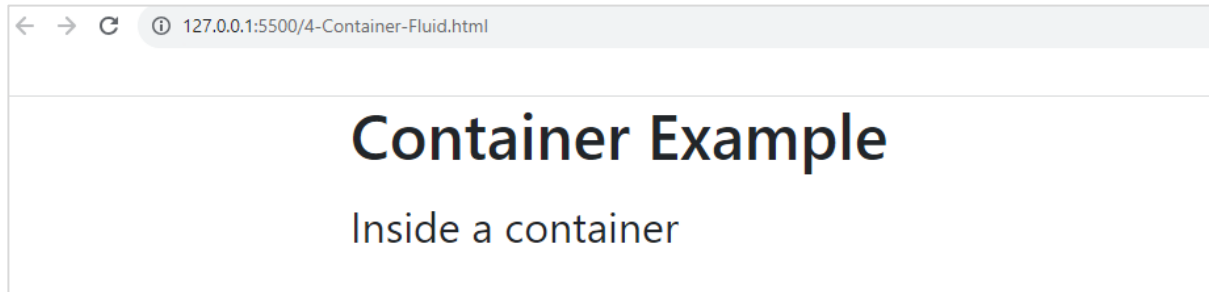
```
<div id="myCarousel" class="carousel slide"
  <!-- Indicators -->
  <ul class="carousel-indicators">
    <li data-target="#myCarousel" data-slide
    <li data-target="#myCarousel" data-slide
    <li data-target="#myCarousel" data-slide
  </ul>

  <!-- Slides -->
  <div class="carousel-inner">
    <div class="carousel-item active">
      
    </div>
    <div class="carousel-item">
      
    </div>
  </div>
```

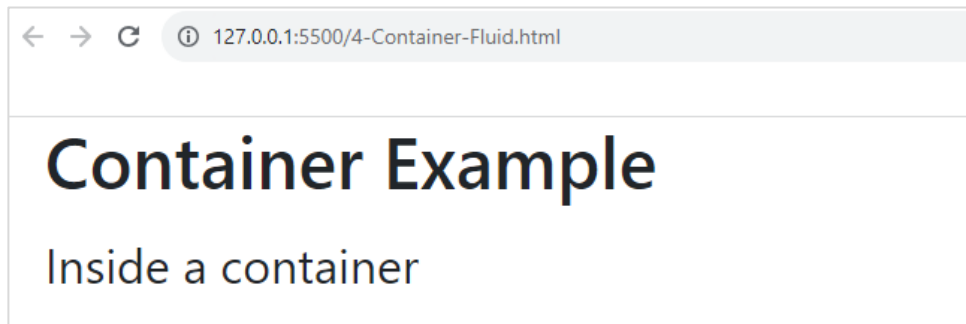
Q. Explain the difference between Bootstrap's **container** and **container-fluid**?



- ❖ .container class **centered the content** and add padding to the sides, ensuring that the content remains within a specified width.
- ❖ .container-fluid class provides a **full-width layout**, with the content extending to the edges of the screen.



```
<body>
  <div class="container">
    <h4>Container Example</h4>
    <p>Inside a container</p>
  </div>
</body>
```



```
<body>
  <div class="container-fluid">
    <h4>Container Example</h4>
    <p>Inside a container</p>
  </div>
</body>
```

**Your fate is in your hands, not in the
hands of your job interviewer.**

Good Luck.

